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MILITARY SPECIFICATION

OSCILLOSCOPE, 5-MEGAHERTZ, MINIATURE, TRANSISTORIZED AN/USM-117( )

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification describes a transistorized general purpose portable oscilloscope for use in observing electrical waveforms and in measuring their voltage and time characteristics.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-T-27 - Transformers and Inductors (Audio, Power and High Power Pulse) General Specification for
MIL-S-901 - Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment and Systems, Requirements for
MIL-E-15090 - Enamel, Equipment, Light-Gray (Formula No. 111)
MIL-E-16400 - Electronic Equipment, Naval Ship and Shore: General Specification
MIL-E-17555 - Electronic and Electrical Equipment, Accessories and Repair Parts; Packaging and Packing of
MIL-T-21038 - Transformer, Pulse, Low Power, General Specification for
MIL-I-45208 - Inspection System Requirements

STANDARDS

FEDERAL

FEDERAL TEST METHOD STANDARD NO. 151 - Metals; Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-200 - Electron Tubes; Selection and Use of
MIL-STD-415 - Test Points and Test Facilities for Electronic Systems and Associated Equipment, Design Standard for
MIL-STD-454 - Standard General Requirements for Electronic Equipment
MIL-STD-461 - Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of
MIL-STD-701 - List of Standard Semiconductor Devices
MS 14009 - Cable Assembly, Power, Electrical

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DRAWINGS

MILITARY

REC 49028	-	UG-625C/U	- Receptacle, Connector, Series BNC
REC 49052	-	UG-274A/U	- Adapter Tee
REB 49064	-	UG-88C/U	- Connector, Plug, for Use with RG 55/U and RG 58/U Cables
RED 2687185	-	UG 255/U	- Adapter BNC to UHF
REC 49438	-	UG 273/U	- Adapter UHF Receptacle to BNC Plug

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification.- Oscilloscopes furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.4).

3.2 Functional description.- This equipment shall provide a visual display of the wave forms of a wide variety of electrical signals. The observed electrical signals shall have a frequency range of dc to 5 megahertz (MHz) and amplitudes of 0.01 volt to 600 volts peak to peak. The equipment shall be so calibrated as to measure the characteristics of waveforms, both amplitude and shape, and shall be suitable for shipboard use.

3.3 Composition.- This equipment shall consist of the following items and other component parts specified herein and as required to make up a complete equipment:

(a) Combination case (see 3.6.1.6).
(b) Panel (see 3.6.1.8).
(c) Vertical channel (see 3.4.3).
(d) Cathode ray tube (see 3.4.2).
(e) High-gain plug-in preamplifier (see 3.4.3.2) MX-2996( )/USM-117.
(f) Dual-channel plug-in preamplifier (see 3.4.3.3) MX-2995( )/USM-117 (when required).
(g) Horizontal channel (see 3.4.4).
(h) Sweep circuits (see 3.4.4.1.6).
(i) Calibrator (see 3.4.5).
(j) Accessories (see 3.4.6).
(k) Outputs (see 3.4.7).
(l) Operating controls (see 3.6.1.7).
(m) Bezel (see 3.6.1.9).

3.4 Performance requirements.-

3.4.1 Warm-up.- The equipment shall be fully operable within 5 minutes after the main power is turned on.

3.4.2 Cathode ray tube.- The wave forms to be examined shall be displayed on a cathode ray tube selected to meet the requirements of this specification. A phosphor P31 screen shall be provided, unless otherwise specified in the contract or order. Other phosphors may be specified (see 6.2). The use of a rectangular cathode ray tube is suggested.

3.4.2.1 Waveform viewing area.- The workable viewing area of the cathode ray tube shall be at least 2-1/2 inches wide by 2-inches high.

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3.4.2.2 Intensity.- The intensity shall be such that a 4-microsecond (μsec) pulse with a rise time of 0.1 μsec or less at a 10-pulse per second (pps) repetition rate and a 5-division amplitude shall be visible. The intensity control circuit shall permit adjustment to provide both greater than and less than optimum intensity.

3.4.2.3 Focus.- The focus circuitry shall be such that when the focus control is adjusted for optimum focus of a spot at the center of the workable area of the cathode ray tube face, the elongation of the spot at any point within the workable area shall not be greater than twice the diameter of the original spot. The spot diameter shall not exceed 1/16 inch when properly focused.

3.4.2.3.1 Astigmatism.- A front panel control and associated circuitry shall be provided which, in conjunction with the focus control, shall permit sufficient adjustment of the spot shape to insure conformance with the focus requirements of 3.4.2.3.

3.4.2.4 Direct access.- Means shall be provided for direct access to the deflection plates, gate output and fuses through a dripproof access cover located on the top of the combination case.

3.4.2.5 Filter.- A compatible filter shall be provided over the cathode ray tube. This filter shall be of an approved design and material selected for optimum viewing of the displayed signal on the type phosphor specified in the contract or order.

3.4.2.6 Graticule.- A graticule shall be furnished, precision ruled accurately in division squares, 10 divisions wide by 8 divisions high. The central vertical and horizontal lines shall be accurately divided at 0.2 division intervals. Each division shall be 0.25 inch. Variable intensity edge lighting of the graticule shall be provided. Each line of the graticule shall be finely etched into clear plastic to preclude halo or parallax effect.

3.4.3 Vertical channel.- The vertical channel circuit shall provide for the use of the plug-in preamplifier units specified herein.

3.4.3.1 Main vertical amplifier.- The pass band of the main vertical amplifier shall be from dc to higher than 10 mc, for a sine wave input.

3.4.3.1.1 Rise time.- The rise time shall be 0.04 μsec or less when measured between the 10 and 90 percent points of maximum amplitude, when a square wave having a rise time of 0.02 μsec is applied.

3.4.3.1.2 Microphonics.- There shall be no visible effects of microphonics.

3.4.3.1.3 Video delay.- A delay network shall be included to properly delay the signal until the sweep has started and the cathode ray tube is unblanked to permit viewing the leading edge of the signal triggering the sweep.

3.4.3.1.4 Stability.- There shall be no instability caused by the application of any signal within the scope of this specification, including asymmetrical pulses.

3.4.3.2 High-gain preamplifier.- Unless otherwise specified in the contract or order, a plug-in high-gain preamplifier shall be provided in the equipment. All power and internal signal paths shall be automatically connected when the preamplifier is inserted into the oscilloscope.

3.4.3.2.1 Composition.- The high-gain preamplifier shall contain a suitable amplifier and shall include attenuators and an input selector.

3.4.3.2.1.1 Amplifier.- The bandpass of the plug-in preamplifier shall be from dc to a frequency higher than 6 mc, to allow the overall bandpass of the plug-in preamplifier in conjunction with the main vertical amplifier to be 5 mc or higher.

3.4.3.2.1.1.1 Bandwidth.- The bandwidth (including high-gain preamplifier, main vertical amplifier and vertical deflection plates of the cathode ray tube) shall be from dc to 5.0 MHz or higher for a sine wave input.

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Both dc and ac coupling shall be provided. Frequency response shall be measured in dB, with "0" dB referred to the level at 10 kHz. The high frequency end of the passband shall approximate a Gaussian rolloff normalized to a 3 dB frequency of 5 MHz or greater (0 dB at 0.2 f_3dB, -0.28 dB at -3.0 f_3dB, -0.75 dB at 0.5 f_3dB, -1.9 dB at 0.8 f_3dB, 3.0 dB at f_3dB, -4.3 dB at 1.2 f_3dB, -5.9 dB at 1.4 f_3dB, -8.6 dB at 1.7 f_3dB, -12 dB at 2 f_3dB). The bandpass curve shall be a smooth curve lying between the limit curves of figure 1. From 20 Hz to 0.13 f_3dB the frequency response shall not be down more than 0.5 dB. The response shall not be above the 10 kHz level at any point between 20 Hz and 0.13 f_3dB. There shall be no positive slope of the response curve about 10 kHz. For dc coupling the response shall meet the requirements for ac coupling for frequencies greater than 20 Hz. From dc to 20 Hz the response shall be down not more than 0.5 dB. These response criteria shall be maintained for all settings of the sensitivity controls.

3.4.3.2.1.1.2 Square wave response.- The square wave response of both the plug-in preamplifier and the main vertical amplifier shall permit faithful reproduction of square waves having a 0.05 µsec rise time over the frequency range of 1 Hz to 250 kHz. There shall be less than 2 percent of overshoot ringing, slope droop, or spurious responses on the displayed square waves.

3.4.3.2.1.1.2.1 Rise time.- The rise time shall be such as to permit an overall (plug-in and main amplifier) vertical channel rise time of less than 0.07 µsec.

3.4.3.2.1.1.3 Sensitivity.- The sensitivity shall be from 0.01 volt per division to 50 volts per division for an ac signal; from 0.1 volt per division to 50 volts per division for a dc signal. Input signal voltages up to 600 volts shall not damage the equipment.

3.4.3.2.1.1.4 Input impedance.- The signal input impedance shall be within ±3 percent of 1 megohm shunted by an effective capacitance of not more than 47 micromicrofarads (pF). The input capacitor for ac coupling shall be 0.1 µf.

3.4.3.2.1.2 Attenuators.- The attenuator controls and circuitry shall provide 11 calibrated sensitivity ranges in 1-2-5-10 sequence from 0.01 volt per division to 20 volts per division and a continuous uncalibrated vernier adjustment between ranges and up to 50 volts per division. Impedance shall be constant within ±3 percent for all settings.

3.4.3.2.1.3 Vertical positioning.- A vertical positioning control shall have a wider range than required for the vertical positioning of any signal within the range of this specification at any point within the working area of the cathode ray tube.

3.4.3.2.1.4 Input selector.- An input selector shall be provided for selection of ac or dc coupling or both inputs.

3.4.3.2.2 Voltage accuracy and linearity.- When the equipment is calibrated with the attenuator controls set for maximum sensitivity, the accuracy on all other calibrated ranges for any trace deflection within the working area of the cathode ray tube face shall be 5 percent or better except that the accuracy shall be 10 percent or better over temperature and humidity range III of 3.7.2.

3.4.3.2.3 Isolation.- There shall be at least 50 dB isolation between the two inputs under any possible combination of controls and input signals within the ranges of this specification.

3.4.3.3 Dual-channel preamplifier.- If required (see 6.2), a plug-in dual-channel preamplifier shall be provided in lieu of (or in addition to) the high-gain preamplifier. This preamplifier shall be mechanically and electrically interchangeable with the high-gain preamplifier without any internal adjustments. All power and internal signal paths shall be automatically connected when the preamplifier is inserted into the oscilloscope.

3.4.3.3.1 Composition.- The dual-channel preamplifier shall contain two identical amplifier circuits and shall include a vertical presentation mode selector circuit, attenuators, polarity invertors, input selectors, and vertical positioning circuitry.

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3.4.3.3.1.1 Mode selector circuit.- The mode selector circuit shall provide for:

(a) The display of either of the two independent channels.
(b) "Alternate" display whereby one sweep causes the display of the input to one channel and the next causes the display of the input to the other channel.
(c) "Chopped" display whereby the sweep is alternately switched between the outputs of the two channels by an electronic switching circuit operating at a rate of 100 kilohertz (kHz) ±10 percent. Provision shall be made for blanking out the switching transients when the "chopped" mode is being used.
(d) A "differential" display of the output of one channel minus the output of the other channel. This may be accomplished by use of a polarity inverter switch.
(e) A "differential" (algebraic sum) display of the output of one channel plus the output of the other channel.

3.4.3.3.1.2 Amplifiers.- The bandpass of each of the two identical amplifiers of the dual-channel plug-in preamplifier shall be from dc to a frequency higher than 6 MHz, to allow the overall bandpass of the plug-in preamplifier in conjunction with the main vertical amplifier to be 5 MHz or higher.

3.4.3.3.1.2.1 Bandwidth.- The bandwidth (including the dual channel preamplifier, main vertical amplifier, and vertical plates of the cathode ray tube) shall be from dc to 5.0 MHz or higher for sine wave input. Both dc and ac coupling shall be provided. Frequency response shall be measured in dB, with "0" dB referred to the level at 10 kHz. The high frequency end of the passband shall approximate a Gaussian rolloff normalized to a 3 dB frequency of 5 MHz or greater (0 dB at 0.1 f_3dB, -0.18 dB at -3.0 f_3dB, -0.75 dB at 0.5 f_3dB, -1.9 dB at 0.8 f_3dB, 3.0 dB at f_3dB, -4.3 dB at 1.2 f_3dB, -5.9 dB at 1.4 f_3dB, -8.6 dB at 1.7 f_3dB, -12 dB at 2 f_3dB). The bandpass curve shall be a smooth curve lying between the limit curves of figure 1. From 20 Hz to 0.13 f_3dB the frequency response shall not be down more than 0.5 dB. The response shall not be above the 10 kHz level at any point between 20 Hz and 0.13 f_3dB. There shall be no positive slope of the response curve above 10 kHz For dc coupling the response shall meet the requirements for ac coupling for frequencies greater than 20 Hz. From dc to 20 Hz the response shall be down not more than 0.5 dB. These response criteria shall be maintained for all settings of the sensitivity controls.

3.4.3.3.1.2.2 Square wave response.- The square wave response of both the plug-in preamplifier and the main vertical amplifier shall permit faithful reproduction of square waves having a 0.05 µsec rise time over the frequency range of 1 Hz to 250 kHz. There shall be less than 2 percent of overshoot, ringing, slope droop, or spurious responses on the displayed square waves.

3.4.3.3.1.2.2.1 Rise time.- The rise time shall be such as to permit an overall (plug-in and main amplifier) vertical channel rise time of less than 0.07 µsec.

3.4.3.3.1.2.3 Sensitivity.- The sensitivity range shall be from 0.05 volt per division to 50 volts per division for both ac and dc coupling. Input signal voltages up to 600 volts shall not damage the equipment.

3.4.3.3.1.2.4 Input impedance.- The signal input impedance shall be within ±3 percent of 1 megohm shunted by an effective capacitance of not more than 30 pF. The input capacitor for ac coupling shall be 0.1 µf.

3.4.3.3.1.3 Attenuators.- Separate attenuator controls and circuitry for each channel shall provide 10 calibrated sensitivity ranges in 1-2-5-10 sequence from 0.05 volt per division to 20 volts per division and a continuous uncalibrated vernier adjustment between ranges and up to 50 volts per division. Impedance shall be constant within ±3 percent for all settings.

3.4.3.3.1.4 Polarity inverter.- Separate polarity inverter switches and circuitry for each channel shall be provided to invert each of the displayed signals.

3.4.3.3.1.5 Vertical positioning.- Separate controls for each channel shall have a wider range than required for the vertical positioning of any signal within the ranges of this specification at any point within the working area of the cathode ray tube.

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3.4.3.3.1.6 Input selector.- An input selector shall be provided for selection of ac or dc coupling for both inputs.

3.4.3.3.2 Voltage accuracy and linearity.- When either channel is calibrated with its attenuator controls set for maximum sensitivity, the accuracy on all other calibrated ranges of that channel for any trace deflection within the working area of the cathode ray tube face shall be 5 percent or better, except that the accuracy shall be 10 percent or better over temperature and humidity range III of 3.7.2.

3.4.3.3.3 Isolation.- There shall be at least 50 dB isolation between the two channels under any possible combination of attenuator controls and input signals within the ranges of this specification.

3.4.4 Horizontal channel.- The horizontal channel shall be capable of providing for the display of an external signal or of providing for the horizontal deflection as a function of time of the signal applied to the vertical channel input.

3.4.4.1 Composition.- The horizontal channel shall contain a horizontal amplifier, a horizontal position control, attenuators, input selectors, a sweep generator and sweep gate, a sweep triggering circuit, sweep mode, blanking and a sweep expanding circuit.

3.4.4.1.1 Horizontal amplifier.- The horizontal amplifier shall consist of a main horizontal amplifier which drives the deflection plates and a preamplifier for external signals which contains an attenuator and a vernier gain control. The deflection shall be to the right for positive input signals. The horizontal amplifier shall meet the requirements specified herein.

3.4.4.1.1.1 Bandwidth.- The bandwidth shall be from dc to 500 kHz or higher for a sine wave input. The frequency response shall be measured in dB, with 0 dB referred to the level at 10 kHz. From 0.1 f_3dB the response shall approximate a Gaussian rolloff normalized to a 3 dB frequency of greater than 500 kHz (0 dB at 0.1 f_3dB, -0.18 dB at 0.3 f_3dB, -0.75 dB at 0.5 f_3dB, -1.9 dB at 0.8 f_3dB, -3 dB at f_3dB, -4.3 dB at 1.2 f_3dB, -5.9 dB at 1.4 f_3dB, -8.6 dB at 1.7 f_3dB, -12 dB at 2.0 f_3dB). There shall be no positive slope of the response curve from dc to f_3dB and shall be down not more than 0.5 dB of the 10 kHz level from dc to 0.1 f_3dB. This response shall be maintained for all settings of the sensitivity controls.

3.4.4.1.1.2 Square wave response.- The square wave response shall permit correct reproduction of 1 Hz to 25 kHz, square waves whose rise time measured between the 10 and 90 percent amplitude points is 0.7 µsec or less. Square waves in the same frequency range having rise times as small as 0.25 µsec shall not be deteriorated except that the displayed rise time measured between the 10 and 90 percent amplitude points shall not exceed 0.74 µsec. Ringing, overshoot, notches, slope droop and spurious responses shall not exceed ±2 percent of the maximum amplitude of the displayed signal.

3.4.4.1.1.3 Sensitivity.- The sensitivity shall be from 0.5 volt per division to 5 volts per division. Input signals up to 600 volts shall not damage the equipment.

3.4.4.1.1.4 Input impedance.- The signal input impedance shall be at least 100 kilohms shunted by not more than 30 pF.

3.4.4.1.2 Attenuators.- The attenuator control and circuitry shall provide one calibrated sensitivity range of 0.5 volt per division and an uncalibrated vernier adjustment up to 5 volts per division. Additional calibrated sensitivity ranges may be provided at the supplier's option providing such ranges are selected from a 1-2-5-10 sequence.

3.4.4.1.3 Horizontal positioning.- A horizontal positioning control shall provide for the positioning of the horizontal trace so that the start of the sweep occurs at any desired position between the center and the left side of the working area of the cathode ray tube. When used with the sweep expander, it shall permit any portion of the trace to be centered on the cathode ray tube.

3.4.4.1.4 Voltage accuracy and linearity.- When calibrated at maximum sensitivity, the accuracy on all the other calibrated ranges for any trace deflections within the working area of the cathode ray tube face shall be 5 percent or better, except that the accuracy shall be 10 percent over temperature and humidity range III as specified in 3.7.2.

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3.4.4.1.5 Isolation.- There shall be at least 50 dB isolation between the vertical and horizontal channels for all possible combinations of attenuator controls and input signals within the ranges of this specification.

3.4.4.1.6 Horizontal sweep.- The horizontal sweep circuitry shall provide either triggered or recurrent linear sweeps at the choice of the operator. When no signal is being fed into the vertical amplifier input, an internally generated signal shall cause a straight horizontal trace to appear on the face of the cathode ray tube.

3.4.4.1.6.1 Sweep speed.- The calibrated sweep speed shall be from 0.1 μsec per division to 0.1 second per division.

3.4.4.1.6.2 Sweep calibration.- A single control shall provide 19 calibrated sweep speed ranges in 1-2-5-10 sequence from 0.1 μsec per division to 0.1 second per division. An uncalibrated vernier control shall provide for continuous adjustment of the sweep time between calibrated ranges. The accuracy of the indicated calibrated sweep speed shall be 3 percent, except that the accuracy shall be 5 percent over temperature and humidity range III as specified in 3.7.2.

3.4.4.1.6.3 Sweep linearity.- The time represented by any division of the normal unexpanded trace shall be within ±3 percent of the time represented by the lefthand (beginning) division of the trace over temperature and humidity range I, 5 percent over temperature and humidity range II, and 8 percent over temperature and humidity range III of 3.7.2.

3.4.4.1.7 Sweep expansion.- A single control shall provide for a five times expansion about the center of any part of the displayed vertical input signal, thus giving an effective sweep length of 50 divisions. The portion of the sweep to be expanded shall be selected by the horizontal positioning control. This expansion shall not add any inaccuracy beyond that of 3.4.4.1.6.2, except for sweep speeds less than 0.1 μsec per division, which shall have an accuracy of 10 percent over temperature and humidity ranges I and II of 3.7.2 and 15 percent over temperature and humidity range III of 3.7.2.

3.4.4.1.8 Blanking.- Provision shall be made for blanking out the return trace of the horizontal sweep.

3.4.4.1.9 Jitter.- There shall be no observable jitter in the horizontal sweep.

3.4.4.1.10 Sweep mode.- The following sweep modes shall be selected by a suitable control:

(a) Preset.
(b) Triggered.
(c) Free running.

3.4.4.1.10.1 Preset mode.- The preset mode shall provide for automatic triggering by common waveforms up to at least 1 MHz.

3.4.4.1.11 Sweep trigger circuit.-

3.4.4.1.11.1 Trigger modes.- The sweep circuit shall permit triggering by any one of the following signals:

(a) Internal vertical amplifier signal giving at least one-half division vertical deflection on the cathode ray tube.
(b) Internal 6.3 volt source of line frequency.
(c) External signal, either dc or ac coupled, of at least one-half volt peak to peak.

3.4.4.1.11.2 Trigger level.- The triggering level shall be continuously adjustable from -30 volts to +30 volts for external trigger signals and to any visible rising point on the wave form for internal signals.

3.4.4.1.11.3 Trigger slope.- The polarity of the triggering signal shall be selected by a trigger slope control.

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3.4.4.1.11.4 Input impedance.- The external trigger input impedance shall be at least 100 kilohm ±10 percent shunted by not more than 100 pF.

3.4.5 Calibrator.- A voltage calibrator meeting the requirements specified herein shall be provided.

3.4.5.1 Output.- A single control shall provide calibrated voltages of 0.04 volt and 0.4 volt.

3.4.5.1.1 Voltage accuracy.- The amplitude of the calibrator voltage shall be within 2 percent of the specified value, except that the accuracy shall be 3 percent over temperature and humidity range III of 3.7.2.

3.4.5.2 Waveshape.- The calibrating voltage shall be a square wave.

3.4.5.2.1 Rise time and fall time.- The rise time and fall time of the calibrator square waves shall each be less than 1.5 μsec measured between the 10 and 90 percent amplitude points.

3.4.5.3 Frequency.- The frequency of the calibrator voltage shall be 1,000 Hz ±10 percent.

3.4.6 Accessories.- The accessories specified herein shall be included.

3.4.6.1 Power cable assembly.- The power cable assembly shall be type A in accordance with MS 14009.

3.4.6.2 Rf probes.- Two high impedance probes each consisting of approximately 54 inches of nonringing, low-noise cable terminated at one end with a male BNC connector and terminated at the other end with a probe presenting an impedance of 10 megohms ±10 percent shunted by 13 pF when used with this oscilloscope. Each probe shall be provided with at least the following tips: flattened alligator jaw, pincer jaw, banana tip, pin tip, hook tip, and spring tip. These rf probes shall be Hewlett-Packard type E01-10001A or Hickok 1804-19, General Atronics Type GA-107 PR, or equal, and approved by the Command or Agency concerned.

3.4.6.3 Rf cables.- Two rf cables consisting of 8 feet of type RG-58C/U cable terminated at each end with a BNC connector type UG-88C/U in accordance with REB49064 shall be furnished.

3.4.6.4 Adapters and tee connectors. - The adapters and connectors specified herein shall be provided.

3.4.6.4.1 Adapters.

(a) Two BNC male to UHF female type UG-255/U in accordance with RED2687185.
(b) Two BNC female to UHF male type UG-273/U in accordance with REC49438.
(c) Two adapters type UG-1441/U or type UG-1035U, binding posts connected to BNC male connectors.

3.4.6.4.2 Tee connectors. - Two BNC tee connectors type UG-274A/U in accordance with REC49052 shall be furnished.

3.4.6.5 Plug-in test cable. - A test cable assembly, at least 1 foot in length, shall be provided to enable the plug-in units to operate outside of the equipment.

3.4.7 Outputs. - The following outputs shall be provided:

(a) Calibration voltages as specified in 3.4.5.
(b) Gate voltage output - a 20-volt pulse available for the duration of the sweep capable of working into 100 kilohms and 60 pF.

3.4.8 Power supply. - The internal electronic power supply shall conform to the following requirements.

3.4.8.1 Regulation. - All dc voltages and currents furnished by the power supply shall be electronically regulated. The degree of regulations shall be sufficient to allow the equipment to function within the requirements of this specification and to ensure that no noticeable changes in the output presentation occur when the power input voltage (including transients) is within the range specified in 3.5.1.

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3.4.8.2 Ripple.- The ac ripple components of regulated dc voltages shall be of such a low magnitude that no undesirable beating or unstable indications shall occur in the output presentation.

3.4.8.3 Ballast tubes.- Ballast tubes shall not be used.

3.4.8.4 Design factor.- In order to provide for future modernization of this equipment all dc voltage or current supplies shall be designed to provide up to 120 percent of the load imposed by this equipment while performing in the optimum manner under the most adverse conditions of this specification.

3.4.8.5 Voltages.- All dc voltages supplied by the power supply shall be as approved by the command or agency concerned, except that the following standard voltages require no further approval:

6.3	45.0	250
12.6	75.0	300
18.9	100	450
25.2	150	1000
	200	

3.4.8.6 Ground.- The design of the power supply unit shall permit optimum performance of the equipment when either or neither side of the power supply is grounded.

3.4.8.7 Transformer hum.- There shall be no audible transformer hum when the equipment is in operation.

3.4.8.8 Contamination.- All power or other frequencies generated in or appearing in the power supply shall be effectively controlled to preclude contamination of any signal or voltage which could result in additional errors in the output presentation.

3.4.8.9 Overload protection.- The power supply shall be protected against overloads caused by failures of other parts by a suitable combination of fuses and current limiting.

3.5 Electrical requirements.-

3.5.1 Power source.- The equipment shall be designed to operate from 115 volts ±10 percent single phase ac at 50 Hz ±5 percent, 60 Hz ±5 percent and 400 Hz ±10 percent when either or neither side of the power input line is grounded.

3.5.1.1 Voltage transients.- The equipment shall be capable of withstanding a voltage transient of ±50 percent from any point within the ±10 percent steady-stage tolerance band of 3.5.1, recoverable to this point within 2 seconds. Momentary impairment of operation during the transient is permissible, but the transient shall not prevent resumption of normal operation.

3.5.1.2 Frequency transients.- The equipment shall be capable of operation during frequency transients of ±3 percent, of which not more than one percent is outside the steady-stage tolerance band of frequency as specified in 3.5.1. The transient shall recover to the steady-state frequency within this band within 2 seconds. Momentary impairment of operation during the transient is permissible, but the transient shall not prevent resumption of normal operation.

3.5.2 Total power.- The total power required to operate the equipment shall not exceed 50 watts.

3.5.3 Electromagnetic interference characteristics.- The equipment shall conform to the applicable electromagnetic interference requirements for class II of MIL-STD-461 (see 4.5.5.3.11). Any parts employed for the reduction of electromagnetic interference, such as capacitors and filters, shall be approved by the command or agency concerned (see 6.3).

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3.5.4 Stability.- There shall be no visible instability in the trace on the cathode ray tube when the power input voltage is varied within the range of 3.5.1 or is subjected to transients within that range.

3.6 Mechanical requirements.-

3.6.1 Construction.- The equipment shall use miniature unitized construction where possible. Optimum use of replaceable subassemblies shall be made.

3.6.1.1 Accessibility.- The construction shall allow maximum accessibility to internal parts and subassemblies.

3.6.1.2 Plug-in vertical preamplifiers.- Provision shall be made to accommodate plug-in vertical preamplifiers of the electrical designs specified herein through an opening in the front of the equipment panel.

3.6.1.3 Finish.- The exterior finish of the equipment shall be gray enamel in accordance with type III, class 2, of MIL-E-15090.

3.6.1.4 Dimensions.- The dimensions of the equipment, excluding the cover, handle, feet, louvers, rear connectors, and latches shall not exceed 8-3/4 inches by 6-27/32 inches high by 14 inches deep behind the front panel. Dimensions including cover, handle, feet and latches shall not exceed 9-3/4 inches wide by 8-7/32 inches high by 20-1/2 inches deep.

3.6.1.4.1 Dust cover.- If required by the procurement document (see 6.2), a dust cover having the following dimensions shall be supplied: 8-3/4 inches wide by 14 inches deep by 6-27/32 inches high. Openings for ventilation and suitable screening for all openings shall be provided.

3.6.1.5 Weight.- The overall weight of the equipment, including accessories, shall be held to a minimum, consistent with the other requirements of this specification, and shall not exceed 26 pounds.

3.6.1.6 Combination case.- The equipment shall be enclosed in a dripproof aluminum combination case equipped with one removable handle on the top of the case, 4 removable rubber bumper feet on both the rear and the bottom of the case, and two removable latches on each of two opposing sides.

3.6.1.6.1 Accessory stowage.- Provision shall be provided for mounting the technical manual and the accessories except the plug-in test cable (see 3.4.6.5) within the cover.

3.6.1.6.2 Access compartment.- A dripproof access compartment shall be provided to afford access to the following:

(a) Direct connections to the plates of the cathode ray tube.
(b) Positive gate output.
(c) Fuses.
(d) CRT positioning lever.

3.6.1.7 Controls.- All controls necessary for operating the equipment shall be placed on the front panel except as noted in 3.6.1.6.2. If internal-external concentric controls are used, the internal knobs shall be of such a design that it will be possible to turn them while wearing gloves without turning the external knob.

3.6.1.8 Panel.- The layout of the panel, including the plug-in unit panels, shall be as specified by the Command or Agency concerned.

3.6.1.8.1 Panel marking.- Panel markings shall be engraved, cast or etched into the metal. When required by the contract or order, fluorescent filling shall be used in the panel marking and knob pointers to facilitate operation in subdued light.

3.6.1.9 Bezel.- A lightproof bezel shall be provided. This bezel shall be designed to shield effectively the face of the cathode ray tube from overhead light, shall be readily removable for the cathode ray tube replacement, and shall provide for ease of filter replacement.

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3.6.2 Input connections.-

3.6.2.1 High-gain preamplifier.- Two inputs and an input selector switch shall be provided for the high-gain preamplifier. The connectors shall be BNC type 625C/U in accordance with Drawing REC 49028.

3.6.2.2 Dual-channel preamplifier.- A channel A and a channel B input connector shall be provided. Connectors shall be BNC type 625C/U.

3.6.2.3 Horizontal channel.- The input connector to the horizontal channel shall be BNC type 625C/U.

3.6.2.4 Trigger circuit.- The trigger input connector shall be a BNC connector type 625C/U.

3.6.2.5 Z-axis connection.- The input connector for Z-axis modulation shall be BNC type 625C/U.

3.6.3 Output connections.- The following output connectors shall be provided.

3.6.3.1 Calibrator.- The calibrator output connector shall be a BNC connector type 625C/U.

3.6.3.2 Gate.- The sweep gate voltage output connector shall be a BNC connector type 625C/U.

3.7 Environmental requirements.-

3.7.1 Shock, vibration and inclination.- This equipment shall meet the shock, vibration, and inclination requirements of MIL-E-16400, except as specified herein.

3.7.2 Temperature and humidity.- This equipment shall operate within the accuracies specified herein over the entire temperature range from -28° to +65°C, with a relative humidity of 95 percent and shall conform to Class 2 of MIL-E-16400, except as specified herein. Where reference is made to temperature and humidity ranges specified herein, the following shall apply: Range I shall be from +15° to +35°C, with a relative humidity not greater than 75 percent; range II shall be from 0° to +65°C, with a relative humidity not greater than 90 percent; range III shall be from -28° to +65°C, with a relative humidity not greater than 95 percent. Storage temperatures up to 85°C, shall not damage the equipment.

3.7.3 Altitude.- There shall be no degradation of performance when the equipment is subjected to decreased barometric pressures of not less than 3.4 inches of mercury approximating an altitude of 50,000 feet.

3.8 Parts and materials requirements.- Regardless of any other requirements, materials and parts containing mercury shall not be used in this equipment unless specifically required or approved by the command or agency concerned, and unless the presence of mercury is noted in that requirement or approval.

3.8.1 Materials.- The materials used in this equipment shall be selected in accordance with the requirements of MIL-E-16400.

3.8.1.1 Corrosion.- The equipment shall be protected from corrosion in accordance with the requirements of MIL-E-16400.

3.8.1.2 Fungus.- The materials used in this equipment shall be selected in accordance with the fungus inert materials required of MIL-E-16400.

3.8.2 Parts.- Unless otherwise specified herein, all parts used in this equipment shall be selected in accordance with the requirements of MIL-E-16400.

3.8.2.1 Standard parts.- Unless otherwise specified herein or otherwise approved by the command or agency concerned, all parts shall be standard parts as defined in MIL-E-16400.

3.8.2.1.1 Electron tubes and semiconductor devices.- Electron tubes and semiconductor devices used in this equipment shall be selected in accordance with the following documents:

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(a) Electron tubes - MIL-STD-200.
(b) Semiconductors (including transistors and semiconductor diodes) - MIL-STD-701.

3.8.2.1.2 Transformers and inductors.- All audio, power and pulse transformers and inductors shall conform to grade 4 or 5, class R, S, or T, life expectancy X in accordance with MIL-T-27 and class 2 of MIL-E-16400, except that low level pulse transformers shall conform to MIL-T-21038.

3.8.2.1.2.1 Transformer diagrams.- A transformer diagram shall be printed on the case of each transformer, or on a plate attached to the case. All other data pertinent to the transformer type shall be marked on the transformer case as required in MIL-T-27.

3.8.2.2 Nonstandard parts.- Nonstandard parts shall be in accordance with MIL-STD-454.

3.9 Reliability requirements.-

3.9.1 Reliability.- The equipment will be subjected to continuous use for long periods under the varied and severe conditions of Military service without overhaul and with little maintenance. Reliability of operation should be considered of prime importance in the design and manufacture of the equipment. The manufacturer shall employ all methods possible in the process of manufacture which will ensure quality and maximum reliability consistent with the state of the art. In the functional application of parts to equipment circuits, adequate factors of safety shall be provided by suitable de-ratings from the part specification values in order to ensure high equipment reliability under all service conditions. The design shall include all possible features which will result in reliable and stable operation with reduced requirements for adjustment and alignment, reduced frequency of failure, reduced requirements for maintenance, and simplified maintenance, thus reducing requirements for highly skilled maintenance personnel.

3.9.2 Service conditions.- The equipment shall be designed to give optimum performance and reliable service during continuous or intermittent operating periods of at least 200 hours under the conditions specified herein without the necessity for major adjustment or servicing.

3.10 Maintainability.- The equipment shall be constructed to provide ease of maintenance, accessibility, and replacement of all parts and units.

3.10.1 Plug-in construction.- Optimum use shall be made of plug-in construction.

3.10.2 Interchangeability.- Mechanical and electrical interchangeability shall exist between similar assemblies, subassemblies, and replaceable parts, regardless of the manufacturer or supplier. In the design of the equipment, provisions shall be made for design tolerance sufficient to accommodate various articles such as tubes, resistors, and other parts having the limiting dimensions and characteristics set forth in the specifications for the particular part involved, without departure from the specified performance.

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3.11 Safety requirements.- This equipment shall be designed to minimize the danger of shock or injury to operating and servicing personnel. Safety features shall be provided in accordance with the following, in the precedence listed:

(a) This specification.
(b) MIL-E-16400.

3.11.1 Fail safe.- This equipment shall be designed to minimize undesirable secondary hazards resulting from the failure of a part of circuit.

3.11.2 Ground potential.- The design of this equipment shall place all external metal and other conducting materials at ground potential at all times. Where a ground is part of the circuit, an external or interconnecting cable shall include a ground wire terminated at both ends in the same manner as the other conductors. In no case shall the shield be depended upon for a ground conductor, with the exception that the shield of coaxial conductors may be used as a signal ground.

3.11.3 Terminals.- All terminals, including those used for direct connection to the deflection plates shall be properly spaced, insulated, and protected.

3.11.4 Thermal cut-offs.- If needed for optimum safety, a thermal cut-off device shall be provided to guard against excessive heating.

3.11.5 Control positions.- Positions of controls in any possible combination shall not damage the equipment.

3.11.6 Power switch.- One dual switch shall be used to connect or disconnect both sides of the power line from the external source to the equipment.

3.11.7 Line fuses.- Fuses shall be provided in both sides of the incoming power supply line. These fuses shall immediately follow the power switch in the circuit and may be mounted in the access compartment. Spare fuses shall be mounted in suitable clips.

3.12 Workmanship.- Workmanship shall be in accordance with MIL-E-16400.

3.13 Test points and test features.- Test points and test features shall be provided in accordance with MIL-STD-415.

3.14 General requirement.- Unless otherwise specified herein, this equipment shall conform to the requirements of MIL-E-16400.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's quality assurance system.- The contractor shall provide and maintain an effective inspection and quality assurance system acceptable to the Government. The inspection system shall be in accordance with MIL-I-45208. A current written description of the contractor's quality assurance system shall be prepared and submitted to the cognizant Government inspector (see 6.3). The written description shall include flow charts showing complete flow of material from inspection on receipt through all manufacturing processes to final shipment. The various processes and inspection positions shall be clearly identified on the flow charts and cross-referenced to detailed inspection procedures and criteria of the written description. Any changes to the quality assurance system which might affect the degree of assurance required by this specification or other applicable documents shall be submitted to the Government inspector for evaluation prior to use.

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4.1.2 Government verification.- All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written system description are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all requirements of this specification (not excluding any requirements for which detailed tests are not specified herein). Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.2 Classification of inspection.- The method of examination and testing of the equipment fall within the following classifications:

(a) Qualification inspection (see 4.3)
(b) Quality conformance inspection (see 4.5)
    (1) Product inspection (see 4.5.3)
    (2) Product control inspection (see 4.5.4)
    (3) Environmental inspection (see 4.5.5)
(c) Materials inspection (4.6)
(d) Inspection of preparation for delivery (4.7)
(e) Retention of qualification (4.8)

4.3 Qualification inspection.- Qualification inspection shall be conducted at a laboratory satisfactory to the Government. Qualification inspection shall consist of the inspections specified in 4.5.3, 4.5.4 and 4.5.5. Four models, built with approved parts, shall be furnished for qualification purposes.

4.3.1 Three sets of the information on standard and nonstandard parts required by figure 2 and 3.8.2.2 shall be submitted for approval (see 6.3). One set of the parts information shall be on brown line reproducible master or equal. Qualification tests on the product shall not be authorized until the parts data are acceptable.

4.3.2 The supplier shall perform the inspections of 4.5, 4.5.3, and 4.5.4 on all four qualification models. The tests of 4.5.5 shall be performed only on two of the models. The tests of 4.5.5 may be divided between the two models provided that at least the temperature and humidity, altitude and accelerated life tests are run on one of the models and at least shock and vibration tests are run on the other model. The remaining tests of 4.5.5 may be run on either of these two models.

4.4 Qualification provisions.- Qualification shall be accomplished in accordance with "Provisions Governing Qualification SD-6" (see 6.4 and 6.5).

4.5 Quality conformance inspection.- This inspection shall include such visual, electrical, and mechanical examination and testing of the equipment, materials, subassemblies, parts, and accessories (including source items) during the process of manufacture as may be required to assure that the complete equipment conforms to the requirements of this specification.

4.5.1 Test equipment.- The following items of test equipment or units of equal or greater accuracy and range shall be used to conduct the tests of 4.5.3, 4.5.4, and 4.5.5. Use of any equipment in place of that listed herein shall require approval of the command or agency concerned (see 6.3).

Equipment	Suggested types	Alternate types
Voltage calibrator	Hewlett-Packard Model 738AR	Simpson Model 2600 Radio Frequency Labs. Model 829.
Time mark generator	Tektronix type 180A	-------
Noise field and intensity meter	See MIL-STD-461	-------
Electronic counter	Hewlett-Packard Model 523B	-------
Audio generator	Hewlett-Packard Model 202C	-------

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Equipment	Suggested types	Alternate types
Signal generator	Hewlett-Packard Model 650A	General Radio Model 1001A
Square wave generator	Hewlett-Packard Model 211A	Tektronix Type 105
VTVM	AN/USM-116	Hickok Model 1601
		Hewlett-Packard Model 410B with Model 458A Adapter
(Capacitance meter) with short coaxial cable leads terminating in alligator clips		-------
Capacitance meter	Tektronix type 130A	-------
High frequency oscilloscope	AN/USM-105A	Hewlett-Packard Model 160A with 162A plug-in ETC Model K120 with 70A plug-in
		Tektronix type 531 or 535 with plug-in preamplifier type L.
Variable frequency power supply	CML Model 1420 or 1410B	Radio Frequency Labs. Model 500 or 150
Pulse generator	Hewlett-Packard Model 212A	General Radio Model 1217-A
Wattmeter	Hickok Model 900C	-------
Transformer, filament	115Vac:6.3Vac	-------
Capacitor 470 pF	470 pF mica	-------
1:1 isolation transformer	UTC type R-73 or Chicago type 18-100	Triad Type N-54-M
High voltage power supply	Precise Electronics Model 3000	Electronic Tube Corp. type 101
Autotransformer (variac)	General Radio Model W20G 230 volt input (variac)	-------

Environmental equipment

Vibration test machine	See MIL-E-16400
Shock test machine	See MIL-E-16400
Temperature, humidity and altitude chamber(s)	Any chamber(s) capable of performing the applicable environmental tests specified herein.
Fungus incubator	See MIL-E-16400
Tilt table	A table whose top can be horizontal and can be tilted to 75 degrees from horizontal.

Special equipment

Weighing scale	Detecto, or equal.

4.5.2 Facilities.- The supplier shall furnish any additional facilities, equipment and personnel required to assure that the equipment conforms to the requirements of this specification.

4.5.3 Product inspection.- The following product inspection shall be performed on every unit delivered.

4.5.3.1 Surface inspection.- Each production unit shall be examined to assure high quality of:

(a) Workmanship, assembly and fit, mechanical safety and marking.
(b) Materials, parts and finish.
(c) Treatment for prevention of corrosion.
(d) Accessibility of parts for replacement.

4.5.3.2 Specific operating tests.- Each equipment shall be energized and subjected to an operating test to ensure qualitatively the proper functioning of the equipment and all operating controls, and conformance with the safety requirements.

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4.5.3.2.1 Test equipment.- The following items of 4.5.1 shall be used for product inspection:

(a) Voltage calibrator.
(b) Time mark generator.
(c) Square wave generator.
(d) Signal generator.
(e) High frequency oscilloscope.
(f) Pulse generator.
(g) Capacitor, 470 pF.
(h) Audio generator.

4.5.3.2.2 Test conditions.- Unless otherwise specified in the contract or order, all tests shall be performed under the following conditions:

(a) Ambient conditions.- The tests shall be performed at factory or laboratory atmospheric conditions.
(b) Input power.- The input power shall be 115 volts, ±10 percent, 60 Hz ±5 percent ac.
(c) Warmup.- The equipment shall be turned on and permitted to warm up for 5 minutes before any test is conducted.
(d) Control settings.- The operating controls of the units shall be set as follows:

Power	ON
Horizontal mode	SWEEP
Time/div	0.2 ms/div
Variable time/div	CAL
Trigger level	0
Stability	PRESET
Trigger slope	INT+
Horizontal position	Adjust to center display
Focus	Adjust for best display
Intensity	Adjust for normal brightness
Scale illumination	Adjust for best display
Astigmatism	Adjust for best display

(e) The equipment shall be checked with the high-gain plug-in preamplifier in its place unless otherwise specified herein. The operating controls of the plug-in unit shall be set as follows:

Input selector	A-ac
Volts/div	0.01 volt per div
Variable volts/div	CAL
Vertical position	Center trace

4.5.3.2.3 Display controls.- The display controls shall be checked as follows:

(a) Set the HORIZONTAL MODE switch to X1 and the VOLTS/DIV switch to 10. Position the spot at the center of the graticule. Adjust the FOCUS, INTENSITY and ASTIGMATISM controls for a small round spot. The spot shall not exceed 1/16 inch in diameter. Using the VERTICAL and HORIZONTAL positioning controls, move the spot about the workable area of the cathode ray tube and observe the shape of the spot. The elongation of the spot at any point in the workable area of the tube face shall not be greater than twice the diameter of the original centered spot.
(b) Set the operating controls as specified in 4.5.3.2.2 (d) and (e).
(c) Connect the 75-ohm output of the square wave generator to the vertical INPUT of the equipment with a BNC to BNC cable.
(d) Set the frequency of the square wave generator to 1 kHz, and adjust the output to give a 5- to 6-division display.
(e) Adjust the INTENSITY control. Observe that there is sufficient range of adjustment to provide both greater than and less than optimum intensity of the trace.

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(f) Adjust the SCALE LIGHT control. Observe that there is sufficient range of adjustment to provide both greater than and less than optimum lighting of the graticule.
(g) Position the trace in both vertical and horizontal directions and note the workable viewing area of the cathode ray tube. This area shall be at least 2-1/2 inches (10 divisions) wide by 2 inches (8 divisions) high.

4.5.3.2.4 Horizontal channel.- Unless otherwise specified herein, the performance of the horizontal channel shall be measured with the operating controls of the plug-in preamplifier set as outlined in 4.5.3.2.2.

4.5.3.2.4.1 Square wave response.- The square wave response of the horizontal amplifier shall be measured as follows:

(a) Connect the 75-ohm output of the square wave generator to the HORIZ INPUT of the equipment with a BNC to BNC 75-ohm cable. Shunt the 75-ohm output with the 470-pF capacitor to give the generator a 0.1 μsec rise time. Connect the 600-ohm output of the square wave generator to the EXT TRIGGER terminal of the high frequency oscilloscope with a BNC to GR cable. Connect the SWEEP OUT of the high frequency oscilloscope to the vertical INPUT of the unit under test with a BNC to BNC cable with a UG-273/U adapter.
(b) Set the HORIZ MODE switch of the unit under test to X1. Adjust the 75-ohm output of the square wave generator to give 5 divisions of horizontal deflection of the trace. Turn the 600-ohm output on full and set the frequency to 25 kHz. Set the trigger source switch of the high frequency oscilloscope to EXT AC, SLOPE TO +, LEVEL to 0, and SWEEP TIME to 10 μsec cm.
(c) Adjust the vertical channel sensitivity of the equipment until a display of about 2 cycles of square wave is provided. Observe the quality of the waveform. There shall be no noticeable ringing, oscillations, or spurious responses. For the purpose of precise measurement, any effect greater than ±0.1 division will be considered noticeable.
(d) Set the frequency of the square wave generator to 100 kHz. Check this frequency on the electronic counter.
(e) Adjust the 75-ohm output to give 5 divisions of horizontal deflection. Adjust the vertical sensitivity of the equipment to display one cycle of the 100 kHz, square wave in 7 divisions of vertical deflection. The vertical display is now calibrated for 1.43 μsec/div.
(f) Measure the 10 to 90 percent rise time of the displayed square wave. The rise time shall be 0.5 division (0.715 μsec) or less.

4.5.3.2.4.2 Sweep calibration and linearity.- Sweep calibration and linearity of the horizontal sweep circuits shall be measured as follows:

(a) Set the SIGNAL SELECTOR of the time-mark generator to markers, switch on 100 μsec markers and set the TRIGGER RATE to 10 kHz.
(b) Set the TIME/DIV switch of the oscilloscope to 0.1 ms/div and set the TRIGGER SLOPE control to AC+.
(c) Connect the SIGNAL OUT of the time-mark generator to the vertical input of the equipment with a BNC to BNC cable and type UG-273 adapter or equal. Connect the TRIGGER OUT of the time-mark generator to the EXT, TRIGGER INPUT terminal of the equipment.
(d) Adjust the TRIGGER LEVEL and STABILITY controls to obtain a stable sweep. Adjust the vertical sensitivity to obtain 2 to 3 divisions of vertical display of the markers.
(e) Adjust the horizontal position control to align the first marker with first graticule line. Ensure that the other markers approximately coincide with other graticule lines.
(f) The variation from coincidence between the marker pulses and the vertical graticule lines within the 10-division area shall not exceed 3 percent or 0.3 minor division, compared to the two left hand (beginning) divisions. Slight repositioning may be required to complete these tests.

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(g) Conduct similar calibration and linearity tests on all the other positions of the TIME/DIV switch of the equipment except for the 0.1 and 0.2 μsec positions, choosing the appropriate marker intervals and trigger rates from the time-mark generator for these measurements.
(h) Set the SIGNAL SELECTOR of the time-mark generator to the 5 MHz sine wave position.
(i) Adjust the amplitude controls of the equipment for exactly 6 divisions of vertical display. Set the TIME/DIV switch to 0.1 μsec/div.
(j) Position the display so that the 50 percent amplitude rise point of the third cycle corresponds with the intersection of the fifth vertical and center horizontal graticule lines.
(k) The 50 percent amplitude crossings of the center horizontal graticule line shall be within 3 percent or 0.3 minor division between any two alternate graticule lines. Slight repositioning may be required to complete these tests.

4.5.3.2.4.3 Horizontal position.- The horizontal position range shall be measured with control settings and instrument connections as in 4.5.3.2.4.2 (a) through (e).

(a) Adjust the HORIZONTAL position control from one extreme to the other. The start of the trace shall be capable of being moved to or beyond the center of the scale. The point on the trace originally coincident with the eleventh graticule marking shall be capable of being moved to or beyond the center of the scale.

4.5.3.2.4.4 Sweep expansion calibration.- Sweep expansion calibration shall be measured with control settings and instrument connections as in 4.5.3.2.4.2 (a) through (e):

(a) Set the time mark generator controls for 10 ms markers and a 100 Hz trigger rate. Set the TIME/DIV switch of the equipment to 2.0 ms/div and adjust the vertical sensitivity for 2 to 3 divisions of vertical display. With the HORIZONTAL MODE switch set to the SWEEP position, 3 time marks shall appear on the display.
(b) Align the second time mark with the center vertical graticule line. Set the HORIZONTAL MODE switch to the X5 MAG position. The center marker shall remain within plus or minus 2 divisions of the center vertical graticule line. This test may be repeated at other sweep times at the discretion of the Government inspector.
(c) Set the TIME/DIV switch to the 5.0 ms/div position with the HORIZONTAL MODE switch in the X5 MAG position. With the time mark generator controls set for 10 ms markers and a 100 Hz trigger rate, when one time mark is positioned on the left vertical graticule line the next mark shall be within ±1.5 minor divisions of the eleventh vertical graticule line.
(d) Position the trace through its entire range and the time between consecutive markers shall be within ±1.5 minor divisions of 10 divisions.
(e) Set the signal selector switch of the time mark generator to the 5 MHz sine wave position. Adjust the amplitude controls of the equipment for exactly 6 divisions of vertical display, the HORIZONTAL MODE switch to X5 MAG, and the TIME/DIV switch to 0.5 μsec/div. Position the display so the 50 percent amplitude rise point of the fourth full cycle corresponds with the fifth vertical and the center horizontal graticule lines. The 50 percent amplitude points of the other cycles in the workable area of the cathode ray tube shall cross the center horizontal graticule line within 3 percent or 1.5 minor divisions of the first, third, seventh, ninth, and eleventh vertical graticule lines.
(f) The same point on each subsequent cycle of the display shall cross the same horizontal line on every other vertical graticule line through a total of 8 divisions within ±3 percent. Slight repositioning may be required to complete this test.
(g) There shall be no appreciable horizontal jitter. For the purpose of a precise measure-

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4.5.3.2.4.5 Trigger level.- Trigger level shall be checked with the following test equipment setup:

(a) Connect the output of the signal generator terminated in 600 ohms to the vertical INPUT of the unit under test with a GR to BNC cable and a BNC tee connector. Connect the EXT TRIGGER INPUT terminal of the equipment to the BNC tee connector with a BNC to BNC cable.
(b) Set the frequency of the signal generator to 1 MHz.
(c) Set the vertical VOLTS/DIV switch to the 0.1 position and adjust the output of the signal generator to give about 1 division of vertical deflection. Adjust the TRIGGER LEVEL control of the oscilloscope to obtain a triggered sweep. Reduce the output of the signal generator and simultaneously adjust the TRIGGER LEVEL control to maintain a triggered sweep. Continue until the smallest deflection is reached that will trigger the sweep. Record this vertical deflection. It shall be 0.5 division or less.
(d) Set the TRIGGER SLOPE control of the equipment to AC+. Set the vertical sensitivity to 0.5 volt/div.
(e) Repeat measurement in (c) starting with about 1V of signal as read on the display. The minimum signal to trigger shall be 0.5V peak-to-peak or less.
(f) Set the TRIGGER SLOPE control to DC+ and repeat (e).
(g) Set the VOLTS/DIV control of the equipment to 5 volts/div and the TRIGGER SLOPE control to line.
(h) Connect the input to the filament transformer to any source of approximately 115 Vac 60 cycle. Load the output to give approximately 6.3 Vac and connect to the vertical INPUT of the equipment.
(i) The sweep should be triggered and the presentation stable. Adjustment of the TRIGGER LEVEL control may be necessary.

4.5.3.2.4.6 STABILITY control.- The STABILITY control shall be checked as follows:

(a) No test equipment is needed.
(b) Turn the STABILITY control of the equipment to the extreme of the FREE RUN range. The sweep shall be free running. Turn the STABILITY control all the way in the opposite direction but not into the PRESET position. The sweep shall not operate.
(c) The PRESET position of the STABILITY control shall have been adequate for all previous tests using frequencies under 1 MHz.

4.5.3.2.4.7 Trigger point selection.- Trigger point selection shall be measured as follows:

(a) Connect the output of the audio generator to the vertical INPUT of the plug-in preamplifier with a BNC to GR cable and a BNC tee. Connect the TRIGGER INPUT connector of the equipment to the BNC tee with a BNC to BNC cable.
(b) Set the VOLTS/DIV control of the plug-in preamplifier to 10v/div and set the sweep time to 0.1 ms/div. Set the audio generator to 1,000 cycles and adjust the output of the generator to give 6 divisions deflection. Set TRIGGER SLOPE control to AC+.
(c) Adjust the POSITION control of the plug-in preamplifier to center the trace vertically. Adjust the TRIGGER LEVEL control from one extreme to the other and observe the movement of the starting point of the sweep. The triggering level shall be continuously adjustable to any point on the positive slope of the waveform within the limits of the graticule markings.
(d) Repeat (c) with the TRIGGER SLOPE control set to -AC. The triggering level shall be continuously adjustable to any point on the negative slope of the waveform within the limits of the graticule markings.
(e) Set the TRIGGER SLOPE control of the equipment to DC and repeat (c) and (d). Set the TRIGGER SOURCE control to -DC and repeat (c) and (d).

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4.5.3.2.5 Vertical channel.-

4.5.3.2.5.1 High-gain preamplifier.- The vertical channel shall be checked with the single channel high-gain preamplifier installed. Controls of the main unit of the equipment shall be set as specified in 4.5.3.2.2. The operating controls of the plug-in unit shall be set as specified herein.

Input selector	A-AC
Volts/div	0.01 volt per division
Variable	CAL
Vertical position	Center trace

4.5.3.2.5.1.1 Square wave response and rise time. - The square wave response and rise time shall be checked as specified herein:

(a) Connect the EXTERNAL TRIGGER INPUT of the equipment to the 600-ohm output of the square wave generator using a GR to BNC cable, and advance the 600-ohm output to about halfway.
(b) Set the frequency of the square wave generator to 1 MHz.
(c) Set the TRIGGER SLOPE control of the equipment to -AC and adjust trigger level if necessary to obtain a stable sweep.
(d) Connect the INPUT of the preamplifier to the 75-ohm output of the square wave generator using a 75-ohm BNC to BNC cable. Adjust the 75-ohm output to give 5 divisions of deflection on the display.
(e) Adjust the stability and level controls to obtain a stable triggered sweep.

4.5.3.2.5.1.1.1 Rise time. -

(a) Set the TIME/DIV switch of the equipment to 0.1 usec/div. Set the HORIZONTAL MODE to X5 MAG. Adjust the TRIGGER LEVEL and STABILITY control if necessary to obtain a stable triggered sweep.
(b) Select the leading edge of the second pulse. Position the waveform so the rise time may be measured in the center area of the display. Measure the 10 to 90 percent rise time of the displayed square wave. Since the square wave generator has a rise time of 0.02 μsec, the measured rise time shall not exceed 0.074 μsec.
(c) Repeat steps (a) and (b) with the VOLTS/DIV switch set to 0.1 V/div position.

4.5.3.2.5.1.1.2 Square wave response. -

(a) Set the square wave generator for the frequencies indicated herein. Set the HORIZ MODE control of the equipment to SWEEP, INPUT selector to AC+, and set the TIME/DIV switch as indicated herein. Adjust TRIGGER LEVEL and SWEEP MODE if necessary to obtain a stable display.

Frequency	Time/div
200 kHz	1 μsec/div
30 kHz	10 μsec/div
5 kHz	50 μsec/div

(b) Observe quality of waveform at each frequency. There shall be no noticeable ringing, oscillation, slope droop, or spurious responses. For the purpose of precise measurement any effect greater than ±0.1 division will be considered noticeable.
(c) Set the square wave generator for the frequencies indicated below. Set the HORIZ MODE control of the equipment to SWEEP, the TIME/DIV switch as indicated below, INPUT SELECTOR to DC+ and VOLTS/DIV to 0.1 V/div.

Frequency	Time/div
10 Hz	10 ms/div
1 Hz	0.1 μsec/div

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(d) Make the same observation as (b).
(e) Adjust TRIGGER LEVEL of the unit as far negative as possible while maintaining stable triggering. Use of the SWEEP MODE control may be necessary. The leading edge of the 20 Hz and higher frequency square waves shall be visible.
(f) Reset the INPUT SELECTOR to AC. Reverse the cable connections to the INPUT of the plug-in preamplifier and the EXT TRIGGER INPUT connector of the unit. Set the TIME/DIV switch to 50 μsec/div. Set the frequency of the square wave generator to 5 kHz. Set the 75-ohm output to maximum and adjust the 600-ohm output to give 4 to 5 divisions of deflection on the display.
(g) Adjust sensitivity VARIABLE control from the CAL position to successive points of less sensitivity (at least five points through range of control) and increase the 600-ohm output of the square wave generator to maintain a constant vertical deflection of 5 divisions.
(h) Observe quality of waveform at each point. There shall be no noticeable ringing, oscillations, droop, or spurious responses. For the purpose of precise measurement any effect greater than ±0.1 division will be considered noticeable.

4.5.3.2.5.1.2 Attenuator calibration. - The attenuator calibration shall be measured with the test equipment setup specified herein:

(a) Connect the output of the voltmeter calibration generator to the INPUT of the plug-in preamplifier with a BNC to GR cable. Set the function switch of the voltmeter calibration generator to 400 cycles peak-to-peak.
(b) Set the TIME/DIV control of the equipment to 2 ms/div. Set the vertical sensitivity VARIABLE control to the CAL position. Set the output of the voltmeter calibrator generator to the peak-to-peak setting for the sensitivity to be checked. The peak-to-peak vertical deflection shall be within the specified limits.

Sensitivity (volts per div)	V. cal. gen.	Deflection in divisions
0.01	0.05V	4.8 to 5.2
0.02	0.1 V	4.8 to 5.2
0.05	0.3 V	5.7 to 6.3
0.10	0.5 V	4.8 to 5.2
0.20	1.0 V	4.8 to 5.2
0.50	3.0 V	5.7 to 6.3
1.00	5.0 V	4.8 to 5.2
2.00	10.0 V	4.8 to 5.2
5.00	30.0 V	5.7 to 6.3
10.00	50.0 V	4.8 to 5.2
20.00	100.0 V	4.8 to 5.2
Variable control set for minimum sensitivity	100.0 V	Not more than 2 div

Attenuator calibration shall be checked at intermediate points of the sensitivity control at the discretion of the inspector.

4.5.3.2.5.1.3 Vertical position range. - The vertical position range shall be checked with the same test equipment setup specified in 4.5.3.2.5.1.2.

(a) Set the output of the voltmeter calibration generator to 0.5 V peak-to-peak. Set the VOLTS/DIV control to 0.05V/div and adjust the vertical POSITION control of the plug-in preamplifier from one extreme to the other. The signal shall move beyond the top and bottom graticule markings in both the upward and downward directions.

4.5.3.2.5.2 Dual-channel vertical preamplifier. - When a dual channel preamplifier is required (see 6.2) in lieu of or in addition to the high gain preamplifier, it shall be checked with the preamplifier plugged into the equipment. The operating controls of the plug-in unit shall be set for each channel as specified herein:

Polarity	AC+
Volts/div	0.05 volt per division

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Variable volts/div	CAL
Vertical position	Center trace
Mode	A only

4.5.3.2.5.2.1 Square wave response and rise time. - The square wave response and rise time of channel A shall be measured as follows:

(a) Connect the TRIGGER INPUT of the equipment to the 600-ohm output of the square wave generator with a BNC to GR cable and set the 600-ohm output to maximum.
(b) Set the frequency of the square wave generator to 250 kHz.
(c) Set the TRIGGER SOURCE control of the equipment to AC+ and adjust TRIGGER LEVEL if necessary to obtain a stable sweep.
(d) Connect the CHANNEL A INPUT of the preamplifier to the 75-ohm output of the square wave generator. Adjust the 75-ohm output to give 5 divisions of deflection on the display.

4.5.3.2.5.2.1.1 Rise time. -

(a) Set the TIME/DIV switch of the equipment 0.1 μs/div. Set the HORIZ MODE to X5 MAG. Adjust TRIGGER LEVEL and SWEEP MODE if necessary to obtain a stable display.
(b) Measure the 10 to 90 percent rise time of the displayed square wave. Since the square wave generator has a rise time of 0.02 μsec, the measured rise time shall not exceed 0.074 μsec.
(c) Repeat steps (a) and (b) with the VOLTS/DIV switch set to the 0.1V/div and 1.0V/div positions.

4.5.3.2.5.2.1.2 Square wave response. -

(a) With the square wave generator set at 250 kHz, set the HORIZ MODE of the equipment to SWEEP. Set the TIME/DIV switch to 1 μsec/div. Adjust the TRIGGER LEVEL and SWEEP MODE if necessary to obtain a stable display.
(b) Observe quality of waveform. There shall be no noticeable ringing, oscillation, droop, or spurious responses. For the purpose of precise measurement any effect greater than 2 percent will be considered noticeable.
(c) Adjust the square wave generator to 2 Hz. Set the TIME/DIV switch to 1 ms/div.
(d) Inspect waveform for defects noted in (b).
(e) Adjust TRIGGER LEVEL of the unit as far negative as possible while maintaining a stable presentation. Use of the SWEEP MODE control may be necessary. The entire leading edge of the square wave shall be visible.
(f) Reverse the cable connections to the CHANNEL A INPUT of the plug-in preamplifier and the TRIGGER INPUT connector of the unit. Set the TIME/DIV switch to 50 μsec/div. Set the frequency of the square wave generator to 5 kHz. Set the 75-ohm output to maximum and adjust the 600-ohm output to give 5 divisions of deflection on the display.
(g) Adjust channel A sensitivity VARIABLE control from the CAL position to successive points of less sensitivity (at least 5 points through the range of control) and increase the 600-ohm output of the square wave generator simultaneously to maintain a deflection of 5 divisions at each point.
(h) Observe quality of waveform. There shall be no noticeable ringing, oscillations, droop, or spurious responses. Check each position of the sensitivity control. For the purpose of precise measurement, any effect greater than 2 percent will be considered noticeable.

4.5.3.2.5.2.2 Attenuator calibration. - The attenuator calibration of channel A shall be measured with the following test equipment setup:

(a) Connect the output of the voltmeter calibration generator to the channel A input of the plug-in preamplifier with a BNC to GR cable. Set the function switch of the voltmeter calibration generator to 400 cycles peak-to-peak.
(b) Set the TIME/DIV control of the equipment to 1 ms/div. Set the vertical sensitivity VARIABLE control to the CAL position. Set the output of the voltmeter calibration generator to the setting for the sensitivity to be checked and the peak-to-peak vertical deflection shall be within the specified limits.

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Sensitivity (volts per div)	Calibration generator	Deflection
0.05	0.3V	5.7 to 6.3
0.10	0.5V	4.8 to 5.2
0.20	1.0V	4.8 to 5.2
0.50	3.0V	5.7 to 6.3
1.00	5.0V	4.8 to 5.2
2.00	10.0V	4.8 to 5.2
5.00	30.0V	5.7 to 6.3
10.00	50.0V	4.8 to 5.2
20.00	100.0V	4.8 to 5.2
Variable control set minimum sensitivity	100.0V	Not more than 2 div

Attenuator calibration shall be checked at intermediate points of the sensitivity control at the discretion of the inspector.

(c) Set the VOLTS/DIV of the plug-in preamplifier to 0.05V/div, and turn the VARIABLE control full CCW. Set the voltmeter calibration generator to 0.5V. The deflection shall be less than 4 divisions.

4.5.3.2.5.2.3 Vertical position range.- The vertical position range of channel A shall be checked with the same test equipment setup specified in 4.5.3.2.5.1.2:

(a) Set the output of the voltmeter calibration generator to 0.5V peak-to-peak. Set the VOLTS/DIV control to 0.05V/div and adjust the vertical position control of channel A of the plug-in preamplifier from one extreme to the other. The signal should be moved beyond the 4 unit graticule markings in both upward and downward directions.

4.5.3.2.5.2.4 Channel B. - The performance of channel B of the plug-in preamplifier shall be measured with the same procedure as specified in 4.5.3.2.5.2.1 and 4.5.3.2.5.2.2 with the MODE switch set to B only.

4.5.3.2.5.3 Alternate display. - The alternate display shall be checked with the following test equipment setup:

(a) Connect the TRIGGER INPUT of the equipment to the 600-ohm output of the square wave generator with a BNC to GR cable and set the 600-ohm output to maximum.
(b) Set the frequency of the square wave generator to 50 Hz.
(c) Set the TRIGGER SOURCE control of the equipment to AC+ and adjust the TRIGGER LEVEL if necessary to obtain sweep.
(d) Connect the CHANNEL A INPUT of the equipment to the 75-ohm output of the square wave generator with a BNC to BNC cable. Adjust the 75-ohm output to give 5 divisions of deflection of the display.
(e) To measure the alternate display, set the MODE switch of the preamplifier to ALTERNATE and adjust the vertical position controls of either channel A or channel B, as necessary, to obtain two traces on screen.
(f) Set TIME/DIV switch to 50 ms/div. The two traces shall occur alternately.
(g) Adjust TIME/DIV switch up to 0.1 μsec/div. Two traces shall still be present.
(h) This test may be performed at any sweep time between 0.1 μs/div and 50 ms/div at the discretion of the inspector.

4.5.3.2.5.4 Chopped display. - The chopped display shall be checked with the same test equipment set up as specified in 4.5.3.2.4.1.

(a) To measure the chopped display, set the MODE switch of the preamplifier to CHOPPED and adjust the vertical position controls of either channel A or channel B, as necessary, to obtain two traces on screen.
(b) Set TIME/DIV switch to 20 ms/div. The two traces should occur simultaneously.
(c) Set TIME/DIV switch to 0.1 μsec/div. The transitions between the two traces should be blanked at normal intensity levels.

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(d) To show the chopped nature of the display, remove the cables from the vertical INPUT and TRIGGER INPUT terminals, set the TRIGGER SLOPE control of the unit to INT and adjust TRIGGER LEVEL to obtain a picture that shows the chopped nature of display.
(e) Measure and record the time of one complete cycle of chopping action. This time shall be between 9.0 and 11.0 μsec.
(f) This test may be performed at any sweep time between 0.1 μsec and 20 ms/div.

4.5.3.2.5.5 Added display. - The added display shall be checked using the following setup:

(a) Connect the 76-ohm output of the square wave generator to the CHANNEL A INPUT of the plug-in preamplifier with a BNC to BNC cable. Connect the 600-ohm output of the square wave generator to the channel B input of the plug-in preamplifier with a GR to BNC cable.
(b) Set the square wave generator to 1 kHz and with the MODE switch of the plug-in preamplifier set to A only, adjust to give 2 divisions of deflection.
(c) Set the MODE switch of the plug-in preamplifier to B only and adjust to give 3 divisions of deflection.
(d) Set the MODE switch of the plug-in preamplifier to ADDED. The display shall present A plus B and have an amplitude of 5 divisions ±2 percent.

4.5.3.2.5.6 Differential display.- The differential display shall be checked using the following setup:

(a) Connect the 75-ohm output of the square wave generator to the channel A input of the plug-in preamplifier with a BNC to BNC cable. Connect the 600-ohm output of the square wave generator to the channel B input of the plug-in preamplifier with a GR to BNC cable.
(b) Set the square wave generator to 1 kHz and with the MODE switch of the plug-in preamplifier set to A only adjust to give 4 divisions of vertical deflection.
(c) Set the MODE switch of the plug-in preamplifier to B only and adjust to give 3 divisions of deflection.
(d) Set the MODE switch of the plug-in preamplifier to A minus B. The display shall present A minus B and have an amplitude of 1 division ±2.0 percent.
(e) Set the MODE switch of the plug-in preamplifier to A+B. The display shall present A plus B and have an amplitude of 7 divisions ±2.0 percent.
(f) Adjust the channel B VOLTS/DIV to reduce the display deflection to a minimum disregarding pips occuring at the switching points of the square wave.
(g) Set the plug-in preamplifier MODE switch to A−B and measure the deflection. The deflection shall be 4 divisions ±0.2 division.

4.5.3.2.6 Calibrator. - Unless otherwise specified herein, the calibrator shall be tested with operating controls of the plug-in preamplifier set as specified in 4.5.3.2.2.

4.5.3.2.6.1 Voltage accuracy. - The voltage accuracy of the calibrator shall be measured as follows:

(a) Connect vertical INPUT A-AC of the equipment to the output of the voltage calibrator with a BNC to GR cable.
(b) Set the FUNCTION switch of the voltage calibrator to 400 cycles peak-to-peak and the TRACKING switch to 0.4V. Set the VOLTS/DIV switch of the equipment to 0.1V/div.
(c) Determine that the display is exactly 4 divisions. Slight adjustment of the VARIABLE VOLTS control may be required.
(d) Connect the internal calibrator output of the oscilloscope to vertical INPUT B-AC. The amplitude of the displayed waveform shall be within ±0.4 divisions of 4 divisions.
(e) Repeat (a) through (d) for any other outputs of the internal calibrator using the nearest appropriate output of the voltage calibrator to first calibrate the vertical channel and then determine the voltage accuracy of the internal calibrator to the proper vertical division.

4.5.3.2.7 Miscellaneous. - Unless otherwise specified herein, intensity modulation and probes shall be checked with the operating controls of the plug-in preamplifier set as specified in 4.5.3.2.2.

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4.5.3.2.7.1 Intensity modulation. - Intensity modulation shall be checked as specified herein:

(a) Set the VOLT/DIV switch to 20, the TIME/DIV switch to 20 µsec and connect at BNC TEE connector to the vertical input of the equipment. Connect the 600-ohm output of the square wave generator to the BNC TEE connector with a BNC to GR cable.
(b) Set the frequency of the square wave generator to 10 kHz and its amplitude control for maximum signal level. Adjust the vertical sensitivity of the equipment to give from 3 to 6 divisions of deflection and the trigger controls for a stable display.
(c) Connect a BNC to BNC cable between the BNC TEE connector and the Z AXIS INPUT of the equipment. Reduce the intensity of the equipment display trace until the highest intensity segments are equal to the original level. The upper portion of the display shall be extinguished while the lower portion shall be at the original intensity.

4.5.3.2.7.2 Accessory probes. - The probes shall be checked as follows:

(a) Connect one of the accessory probes to the vertical INPUT of the equipment and set the vertical sensitivity to 1V/div. Set the calibrator output control of the high frequency oscilloscope to 50. Connect the accessory probe tip to the calibrator volts out terminal and the ground lead of the probe to the ground terminal of the high frequency oscilloscope.
(b) Adjust sweep controls as necessary to obtain a stable waveform pattern of two or three cycles.
(c) Back off the locking rings on the probe and rotate the rear portion of the probe relative to the front portion.
(d) Observe that the waveform can be changed from one with overshoot to one with undershoot. Adjust the probe until an optimum square wave is presented and then tighten the locking ring to hold this adjustment. The probe is now compensated at 1 kHz.
(e) Verify that the amplitude of the square wave presentation is between 4.8 and 5.2 divisions.
(f) Repeat this procedure with the other probe furnished.

4.5.3.2.7.2.1 Probe input impedance. - The following test shall be performed on the probe provided.

(a) Zero the capacity meter. After compensating the probe, connect the probe to the capacity meter leads.
(b) With the equipment operating, measure the capacity of the probe in all positions of the volts/div switch. The input capacity shall not exceed 13 pF.
(c) Connect the resistance measuring leads of the vacuum tube voltmeter to the probe. Set the INPUT SELECTOR switch of the preamplifier to A-DC.
(d) Set the selector switch of the VTVM to OHMS and measure the input resistance for all dc-coupled sensitivity settings. The resistance shall be 10 megohms ±10 percent.

4.5.4 Product control inspection. - Product control inspection shall be conducted on a sampling basis in accordance with 4.5.4.1. This inspection shall comprise such examination and testing as will prove satisfactory performance through the entire range of operation, detect deterioration of the design or deviations from approved processing of materials, and demonstrate that the equipment will function satisfactorily under adverse conditions. Unless otherwise specified herein, these tests shall be conducted at normal ambient room temperature.

4.5.4.1 Sampling plan. - From each lot which has passed product inspection (see 4.5.3), a random sample of units shall be selected and subjected to product control inspection. Sampling shall be in accordance with MIL-STD-105, General Inspection Level I, and an AQL of 1.5 percent.

4.5.4.2 Rejection. - If an inspection lot is rejected, the supplier may withdraw the lot from further inspection. The supplier may also rework a rejected lot to correct the defects or screen out the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from the new lots and shall not lose their identity. The results of each test shall be compared with the requirements of this specification. Failure to conform to this specification for any test shall be counted as a defect, and the unit shall be rejected. If the number of rejections in any sample exceeds the acceptance number for the sample, the lot represented by the sample shall be rejected.

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4.5.4.3 Test equipment. - The following items of 4.5.1 shall be used for product control inspection.

(a) VTVM (with coaxial T adapter or BNC adapter).
(b) Signal generator.
(c) Capacitance meter.
(d) Square wave generator.
(e) Variable frequency power supply.
(f) Wattmeter.
(g) Capacitor, 470 pF mica.
(h) Audio generator.
(i) High voltage power supply.
(j) Autotransformer.
(k) Voltage calibrator.

4.5.4.4 Specific product control tests. -

4.5.4.4.1 Test conditions. - Unless otherwise specified herein, test conditions shall be the same as 4.5.3.2.2.

4.5.4.4.2 Vertical channel. -

4.5.4.4.2.1 High-gain preamplifier. - Unless otherwise specified herein, for product control testing, the controls of the high-gain preamplifier shall be set as follows:

Input selector	A-AC
Volts/div	0.01 volt per division
Variable volts/div	CAL
Vertical position	Center trace

4.5.4.4.2.1.1 Bandwidth. - The bandwidth shall be measured as follows:

(a) Connect the BNC tee to the INPUT of the plug-in preamplifier. Insert the ac probe of the VTVM into the receptacle of the VTVM adapter and attach the adapter to the BNC tee with a short BNC to BNC cable. Using the same length cable with a GR to BNC connector, connect the signal generator to the other branch of the tee. The signal generator shall be terminated with a resistance load of 600 ohms.
(b) Set the VOLTS/DIV switch of the plug-in preamplifier to 0.2V/div. Set the voltmeter selector to AC and the range switch to 1.0V. Set the frequency of the signal generator to 10 kHz. Adjust the signal generator output to give 8 divisions of deflection on the display and record reading of the VTVM. Measure all frequency points above the 10 kHz reference first, then measure the points below 10 kHz. Make the 1 kHz measurement using both the VTVM and the high frequency oscilloscope before making the measurements below 1 kHz.
(c) Adjust the frequency of the signal generator to the frequencies specified below and adjust its output to maintain the original reading on the voltmeter. Observe that the amplitude of the trace falls within the following specified limits and record the deflection for each frequency:

Frequency	Deflection div.
1 kHz	7.8 to 8.0
2 kHz	7.8 to 8.0
4 kHz	7.8 to 8.0
6 kHz	7.8 to 8.0
10 kHz	8.0
20 kHz	7.8 to 8.0
40 kHz	7.8 to 8.0
80 kHz	7.8 to 8.0
100 kHz	7.8 to 8.0

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Frequency (cont'd)	Deflection div. (cont'd)
200 kHz	7.8 to 8.0
400 kHz	7.8 to 8.0
500 kHz	7.8 to 8.0
1 MHz	7.47 to 7.91
3 MHz	7.05 to 7.47
4 MHz	6.43 to 6.81
5 MHz	5.66 to 6.00
7.5 MHz	3.8 to 4.2
10 MHz	1.8 to 2.2

(d) At 1 kHz remove the signal generator from the equipment. Connect the audio generator to the high frequency oscilloscope and the VTVM using a TEE connector. Set the frequency of the audio generator and adjust its output level to the 10 kHz reference level as indicated on the VTVM. Set the high frequency oscilloscope controls as specified herein:

(1) Horizontal display switch	EXT
(2) Stability control	PRESET
(3) Input selector	DC
(4) Volts/cm switch and variable control for exactly 4 cm of vertical display.	

Remove the VTVM and connect the high frequency oscilloscope and the audio generator to the equipment using a BNC-TEE connector.

(e) Adjust the audio generator to obtain the frequencies specified below, adjusting its output level to maintain the vertical display of the high frequency oscilloscope at 4 cm. The amplitude of the display on the equipment shall be within the limits specified below. Record the amplitude for each frequency.

Frequency	Deflection div.
10 kHz	8.0
1 kHz	7.8 to 8.0
800 Hz	7.8 to 8.0
600 Hz	7.8 to 8.0
400 Hz	7.8 to 8.0
200 Hz	7.8 to 8.0
100 Hz	7.8 to 8.0
80 Hz	7.8 to 8.0
60 Hz	7.8 to 8.0
40 Hz	7.8 to 8.0
20 Hz	7.8 to 8.0
10 Hz	7.4 to 7.6
4 Hz	7.2 to 7.4
2.9 Hz	6.6 to 6.9
2 Hz	5.6 to 5.8
1.5 Hz	3.8 to 4.2
1 Hz	1.8 to 2.2

(f) If the 3 dB point occurs at a higher frequency than 5 MHz, the frequencies above 100 kHz shall be relatively raised to preserve the Gaussian roll-off characteristics at the higher frequency response level. If the 3 dB point occurs at a lower frequency than 2 Hz, the frequencies below 10 Hz shall be relatively raised to preserve the Gaussian roll-off characteristics at the lower frequency response level.

(g) The inspector may, at his discretion, increase or decrease the number of frequencies at which the amplitude is recorded, except that the 3 dB, 6 dB, and 12 dB points at each end of the band shall be recorded.

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4.5.4.4.2.1.2 DC sensitivity, voltage accuracy, and linearity.-

(a) Set the INPUT SELECTOR on A-DC. Connect the voltage calibrator to vertical INPUT A. Set the calibrator output on DC+.
(b) Set the voltage calibrator output on zero and position the oscilloscope trace on the lowest graticule line.
(c) Set the VOLTS/DIV switch and the voltage calibrator output as shown below. The trace shall be displaced upward within the indicated number of divisions.

Volts/div.	Calibrator voltage	Deflection div.
0.10	0.50	4.75 to 5.25
0.20	1.0	4.75 to 5.25
0.50	1.0	1.90 to 2.10
	2.0	3.80 to 4.20
	3.0	5.70 to 6.30
1.0	5.0	4.75 to 5.25
2.0	5.0	2.40 to 2.60
	10.0	4.75 to 5.25
5.0	20.0	3.80 to 4.20
10.0	50.0	4.75 to 5.25
20.0	100.0	4.75 to 5.25

(d) Rotate the VARIABLE volts/div control for minimum sensitivity. With a calibrator voltage of 100 volts, deflection shall be 2 divisions or less.
(e) The 0.01 through 0.05V/div range shall be checked, using the 400 cycles peak-to-peak output of the calibration generator. The peak-to-peak deflections shall be within the indicated number of divisions.

Volts/div.	Calibrator voltage	Deflection div.
0.01	0.01	0.95 to 1.05
	0.05	4.75 to 5.25
0.02	0.05	2.40 to 2.60
	0.10	4.75 to 5.25
0.05	0.20	3.80 to 4.20

(f) For production items the inspector may, at his discretion, decrease the number of points at which accuracy is checked provided it is checked at 0.01V/div and 1.0V/div for each sample unit.

4.5.4.4.2.1.3 Input impedance.- With the equipment operating, the input impedance shall be measured as follows:

(a) Connect the capacity meter leads to a GR to BNC adapter and adjust the capacity meter to zero capacity, thus balancing out the capacity of the adapter.
(b) With the leads still connected, connect the adapter to the vertical input of the plug-in preamplifier on all sensitivity positions. The input capacity shall not exceed 47 pF.
(c) Disconnect the capacity meter leads. Connect the resistance measuring leads of the VTVM to the adapter. Set the input selector switch of the plug-in preamplifier to A-DC.
(d) Set the selector switch of the VTVM to OHMS RX100K and measure the input resistance for all dc-coupled sensitivity settings. This resistance shall be 1 megohm ±10 percent.

4.5.4.4.2.1.4 High voltage test.-

(a) Connect the leads of the high voltage power supply to INPUT A-DC.
(b) With the equipment operating, apply 600 volts to the input. No damage to equipment or impairment of operation shall occur for any position of the VOLTS/DIV switch.
(c) Repeat this test for INPUT B-AC.

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4.5.4.4.2.1.5 Isolation between inputs.-

(a) Connect the square wave generator 600-ohm output to input B.
(b) Set the frequency of the square wave generator to 1 kHz and the 600-ohm output to 3 volts.
(c) Set the sensitivity to 0.01V/div. Shield INPUT A with BNC cap.
(d) Set the INPUT SELECTOR switch on A-DC. Adjust trigger controls for a stable trace.
(e) Deflection of the trace shall be less than 1 division.
(f) Repeat test reversing the input connection, the selector switch position, and the BNC cap.

4.5.4.4.2.2 Dual-channel preamplifier.- Unless otherwise specified herein, for product control testing, the controls of the dual-channel preamplifier shall be set as follows:

Polarity	AC
Volts/div	0.05 volt per division
Variable volts/div	CAL
Vertical position	Center trace
Mode	A only

4.5.4.4.2.2.1 Bandwidth.- The bandwidth shall be measured as follows:

(a) Connect the BNC tee to the channel A input of the plug-in preamplifier. Insert the ac probe of the VTVM into the receptacle of the VTVM adapter and attach the adapter to the BNC tee with a BNC to BNC cable. Connect the signal generator to the other branch of the tee.
(b) Set the VOLTS/DIV switch of the plug-in preamplifier to 0.2V/div. Set the voltmeter selector to AC and the range switch to 1.0V. Set the frequency of the signal generator to 10 kHz. Turn the SWEEP MODE control of the equipment to the FREE RUN position. Adjust signal generator output to give 8 divisions of deflection on the display and record reading of the VTVM. Measure all frequency points above the 10 kHz reference first, then measure the points below 10 kHz. Make the 1 kHz measurement using both the VTVM and the high frequency oscilloscope before making the measurements below 1 kHz.
(c) Adjust the frequency of the signal generator to the frequencies specified below and adjust its output to maintain the original reading on the VTVM. The amplitude of the displayed waveform shall fall within the limits specified below. Record the deflection for each frequency.

Frequency	Deflection div.
1 kHz	7.8 to 8.0
2 kHz	7.8 to 8.0
4 kHz	7.8 to 8.0
6 kHz	7.8 to 8.0
10 kHz	8.0
20 kHz	7.8 to 8.0
40 kHz	7.8 to 8.0
80 kHz	7.8 to 8.0
100 kHz	7.8 to 8.0
200 kHz	7.8 to 8.0
400 kHz	7.8 to 8.0
500 kHz	7.8 to 8.0
1 MHz	7.47 to 7.91
3 MHz	7.05 to 7.47
4 MHz	6.43 to 6.81
5 MHz	5.66 to 6.00
7.5 MHz	3.8 to 4.2
10 MHz	1.8 to 2.2

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(d) At 1 kHz remove the signal generator from the equipment. Connect the audio generator to the high frequency oscilloscope and the VTVM using a TEE connector. Set the frequency of the audio generator and adjust its output level to the 10 kHz reference level as indicated on the VTVM. Set the high frequency oscilloscope controls as follows:

(1) Horizontal display switch	EXT
(2) Stability control	PRESET
(3) Input selector	DC
(4) Volts/cm switch and variable control for exactly 4 cm of vertical display.	

Remove the VTVM and connect the high frequency oscilloscope and the audio generator to the equipment using a BNC-TEE connector.

(e) Adjust the audio generator to obtain the frequencies specified below adjusting its output level to maintain the vertical display of the high frequency oscilloscope at 4 cm. The amplitude of the display on the equipment shall be within the limits specified below. Record the amplitude for each frequency.

Frequency	Deflection div.
10 kHz	8.0
1 kHz	7.8 to 8.0
800 Hz	7.8 to 8.0
600 Hz	7.8 to 8.0
400 Hz	7.8 to 8.0
200 Hz	7.8 to 8.0
100 Hz	7.8 to 8.0
80 Hz	7.8 to 8.0
60 Hz	7.8 to 8.0
40 Hz	7.8 to 8.0
20 Hz	7.8 to 8.0
10 Hz	7.4 to 7.6
4 Hz	7.2 to 7.4
2.9 Hz	6.6 to 6.9
2 Hz	5.6 to 5.8
1.5 Hz	3.8 to 4.2
1 Hz	1.8 to 2.2

(f) The Government inspector may, at his discretion, increase or decrease the number of frequencies at which the amplitude is recorded except that the 3 dB, 6 dB, and 12 dB points at each end of the band shall be recorded.

4.5.4.4.2.2.2 DC sensitivity, voltage accuracy, and linearity.-

(a) Connect the voltage calibration to the channel A input. Set the calibrator output on DC+.
(b) Set the voltage calibration output on zero and position the oscilloscope trace on the lowest graticule line.
(c) Set the VOLTS/DIV switch and the voltage calibrator output. The trace shall be displaced upward within the specified number of divisions as follows:

Volts/div.	Calibrator voltage	Deflection div.
0.05	0.20	3.00 to 4.20
0.10	0.50	4.75 to 5.25
0.20	1.0	4.75 to 5.25
0.50	1.0	1.90 to 2.10
	2.0	3.80 to 4.20
	3.0	5.70 to 6.30
1.0	5.0	4.75 to 5.25

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Volts/div. (cont'd)	Calibrator voltage (cont'd)	Deflection div. (cont'd)
2.0	5.0	2.40 to 2.60
	10.0	4.75 to 5.25
5.0	20.0	3.80 to 4.20
10.0	50.0	4.75 to 5.25
20.0	100.0	4.75 to 5.25

(d) Rotate the VARIABLE volts/div control for minimum sensitivity. With a calibration voltage of 100 volts, deflection shall be 2 divisions or less.
(e) For production items the inspector may, at his discretion, decrease the number of points at which accuracy is checked provided it is checked at 0.05V/div and 1.0V/div for each sample unit.

4.5.4.4.2.2.3 Input impedance.- With the equipment operating, the input impedance shall be measured as follows:

(a) Connect the capacity meter leads to a GR to BNC adapter and adjust the capacity meter to zero capacity, thus balancing out the capacity of the adapter.
(b) Measure the input capacity of the plug-in preamplifier on all sensitivity positions. The input capacity shall not exceed 30 pF.
(c) Disconnect the capacity meter leads and connect the resistance-measuring leads of the vacuum tube voltmeter to the adapter. Set the polarity switch on DC+.
(d) Set the selector switch on the VTVM to OHMS RX10K and measure the input resistance on all sensitivity positions. This resistance shall be within 10 percent of 1 megohm.

4.5.4.4.2.2.4 High voltage test.-

(a) Connect the leads of the high voltage power supply to the channel A input.
(b) With the equipment operating, apply 600 volts to the INPUT. No damage to the equipment or impairment of operation shall occur for any position of the VOLTS/DIV switch.

4.5.4.4.2.2.5 Isolation.- Isolation of the channel A input from the horizontal circuits shall be checked as follows:

(a) Connect the square wave generator 600-ohm output to the channel A input with GR to BNC cable.
(b) Set the frequency of the square wave generator to 1 kHz and the 600-ohm output to 3 volts. Set the HORIZ MODE switch to X10. Shield the HORIZ INPUT terminal with a BNC cap.
(c) Horizontal deflection shall be less than 0.1 division for any setting of vertical sensitivity.

4.5.4.4.2.2.6 Isolation between channels.-

(a) Connect the square wave generator 600-ohm output to the channel A input.
(b) Set the frequency of the square wave generator to 1 kHz and the 600-ohm output to 3 volts.
(c) Set the channel B sensitivity to 0.1V/div. Shield the channel B input.
(d) Set the vertical MODE switch on ALTERNATE. Adjust trigger controls for two stable traces.
(e) Deflection of the channel B trace shall be less than 0.1 division for any setting of channel A sensitivity.
(f) Repeat test with the vertical MODE switch on CHOPPED.

4.5.4.4.2.2.7 Channel B.- Repeat the tests of channel A for channel B.

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4.5.4.4.3 Horizontal channel.- Unless otherwise specified herein, the horizontal channel performance shall be measured with the operating controls of the plug-in preamplifier set as specified in 4.5.4.4.2.

4.5.4.4.3.1 Bandwidth.- The bandwidth of the horizontal amplifier shall be checked as hereinafter specified:

(a) Connect the BNC tee to the INPUT of the plug-in preamplifier. Insert the ac probe of the VTVM into the receptacle of the VTVM adapter and attach the adapter to the BNC tee with a BNC to BNC cable. Connect the signal generator to the other branch of the tee. The signal generator shall be terminated with a resistive load of 600 ohms.

(b) Set the HORIZ MODE switch of the equipment to X1. Set the voltmeter selector to AC and the range switch to 3.0V. Set the frequency of the signal generator to 10 kHz. Adjust signal generator output to give 6 divisions of deflection on the display and record reading of the VTVM. Measure all frequency points above the 10 kHz reference first, then measure the points below 10 kHz. Make the 1 kHz measurement using both the VTVM and the high frequency oscilloscope before making the measurements below 1 kHz.

(c) Adjust the frequency of the signal generator to the frequencies specified below and adjust its output to maintain the original reading on the VTVM. The amplitude of the displayed waveform shall fall within the limits hereinafter specified. Record the deflection for each frequency:

Frequency	Deflection div.
2 Hz	5.66 to 6.00
10 Hz	5.66 to 6.00
20 Hz	5.66 to 6.00
50 Hz	5.66 to 6.00
60 Hz	5.66 to 6.00
120 Hz	5.66 to 6.00
400 Hz	5.66 to 6.00
1 kHz	5.66 to 6.00
5 kHz	5.66 to 6.00
10 kHz	6.00
25 kHz	5.66 to 6.00
50 kHz	5.66 to 6.00
100 kHz	5.60 to 5.93
300 kHz	5.23 to 5.60
400 kHz	4.82 to 5.11
500 kHz	4.25 to 4.50
1 MHz	1.51 to 1.60

(d) At 1 kHz remove the signal generator from the equipment. Connect the audio generator to the high frequency oscilloscope and the VTVM, using a tee connector. Set the frequency of the audio generator and adjust its output level to the 10 kHz reference level as indicated on the VTVM. Set the high frequency oscilloscope controls as follows:

(1) Horizontal display switch — EXT
(2) Stability control — PRESET
(3) Input selector — DC
(4) Volts/cm switch and variable control for exactly 4 cm of vertical deflection.

(e) If the 3 dB point occurs at a higher frequency than 500 kHz, the frequencies above 10 kHz shall be relatively raised to preserve the Gaussian roll-off characteristics at the higher frequency response level.

(f) The inspector may, at his discretion, increase or decrease the number of frequencies at which the amplitude is recorded, except that the 3 dB, 6 dB, and 12 dB points at each end of the band shall be recorded.

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4.5.4.4.3.2 Input impedance.- With the equipment operating, the input impedance of the horizontal amplifier shall be checked as follows:

(a) Connect a GR to BNC adapter to the capacity meter terminals and zero meter. Connect the BNC adapter to the HORIZ INPUT terminal of the equipment.
(b) Measure the input capacity of all the HORIZ INPUT positions, using all external settings of the HORIZ MODE control. The input capacity shall not exceed 30 pF.
(c) Connect a GR to BNC adapter to the HORIZ INPUT terminals of the equipment. Connect the resistance measuring leads of the vacuum tube voltmeter to the adapter.
(d) Set the selector switch of the VTVM to RX10K OHMS and measure the input resistance for all HORIZ INPUT positions, using all external settings of the HORIZ MODE control. Resistance shall be 100 kilohms or more.

4.5.4.4.3.3 Isolation.- The isolation between the horizontal and vertical circuits shall be checked as follows:

(a) Connect the square wave generator 600-ohm output to the vertical INPUT of the equipment with a GR to BNC cable.
(b) Set the frequency of the square wave generator to 100 kHz and the 600-ohm output to 3 volts. Set the HORIZ MODE switch to X5. Shield the HORIZ INPUT terminal with a BNC cap.
(c) Horizontal deflection shall be less than 0.1 division for any setting of vertical sensitivity from 0.5 to 20.0V/div.

4.5.4.4.3.4 DC sensitivity.- The dc sensitivity of the horizontal channel shall be tested in the following manner:

(a) With no signal into the vertical channel, place the HORIZ MODE switch in the X1 position and position the spot on the central horizontal graticule line at the extreme left hand vertical graticule line.
(b) Connect the output of the voltage calibrator to the INPUT of the horizontal channel and rotate the HORIZ GAIN control fully clockwise (cw).
(c) Set the voltage calibrator function switch to +dc and its voltage output as hereinafter specified. The spot shall be displaced to the right within the specified number of divisions.

Cal. voltage (volts)	Deflection div.
0.5	0.90 to 1.10
1.0	1.80 to 2.20
2.0	3.60 to 4.40
3.0	5.40 to 6.60
4.0	7.20 to 8.80
5.0	9.00 to greater than 10

(d) Repeat the test for any other calibrated horizontal sensitivity ranges, applying the appropriate multiple of the above calibrator voltage (ex., X2, X5) to give the deflections in (c).
(e) With the HORIZ MODE switch set to X1, set the calibrator voltage to 2.5 volts for a 4.40 to 5.50 division deflection. Demonstrate that the HORIZ MODE X5 setting attenuates the deflection to a 0.9 to 1.1 division deflection.

4.5.4.4.3.5 High voltage test.-

(a) Connect the leads of the high voltage power supply to the HORIZ INPUT.
(b) With the HORIZ MODE switch X5 and the equipment operating, apply 600 volts to the INPUT. No damage to the equipment or impairment of operation shall occur.

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4.5.4.4.3.6 Trigger input impedance.- The trigger input impedance shall be tested in the following manner:

(a) Set the TRIGGER SLOPE control to the DC position.
(b) Connect a GR to BNC adapter to the capacity meter leads and zero the meter. Connect the GR to BNC adapter to the trigger input connector of the equipment.
(c) Measure the input capacity of all the external trigger input positions of the trigger slope control. The capacity shall not exceed 100 pF at any position of the TRIGGER SLOPE control.
(d) With the TRIGGER SLOPE control back in a DC position connect the resistance measuring leads of the VTVM to the trigger input connector of the equipment. The VTVM, set at RX10K, shall indicate 100,000 ohms or higher.

4.5.4.4.4 Calibrator.- Unless otherwise specified herein, the calibrator shall be tested with the operating controls of the plug-in preamplifier set as specified in 4.5.4.4.2.

4.5.4.4.4.1 Waveshape.- The waveshape of the calibrator shall be measured as follows:

(a) Select a convenient vertical sensitivity setting.
(b) Operate the calibrate switch and observe the nature of the waveform.
(c) The calibrator voltage shall be a square wave with a period between 0.9 and 1.1 ms.
(d) Adjust the sweep controls to permit measurement of the 10 to 90 percent rise time of the square wave. Readjust for measurement of the decay time. The measured rise and decay times shall be not more than 1.5 μs.
(e) This test may be repeated at any vertical sensitivity setting at the discretion of the inspector.

4.5.4.4.5 Power supply.- Unless otherwise specified herein, the power supply shall be tested with the operating controls of the plug-in preamplifier set as specified in 4.5.3.2.2.

4.5.4.4.5.1 Voltage regulation and frequency stability.- Voltage regulation and frequency stability shall be tested as follows:

(a) Connect the power cord of the equipment to the variable frequency power supply. Connect the vertical INPUT to the output of the voltmeter calibration generator.
(b) Set the output of the variable frequency power supply at 115V, 60 Hz. Set the vertical sensitivity of the equipment to 0.01V/division and the calibration generator output at 50 millivolts peak to peak.
(c) Measure the vertical amplitude of the display. The height of the sine wave shall be between 4.75 and 5.25 divisions.
(d) Disconnect the calibration generator. Set the TRIGGER SOURCE to LINE. Set the sweep time to 5 ms/division.
(e) Measure any vertical deflection observed. This deflection shall not exceed 0.2 division.
(f) Repeat (b) through (e) for voltages and frequencies from the variable frequency power supply as follows:

Voltages Volts	Frequencies Hz
115	63, 57
103.5	63, 57, 60
126.5	63, 57, 60
103.5	53, 47, 50
115.0	53, 47, 50
126.5	53, 47, 50
103.5	440, 360, 400
115	440, 360, 400
126.5	440, 360, 400

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This test may be performed at other points within the specified ranges at the discretion of the inspector.

4.5.4.4.5.1.1 Frequency transients.- Frequency transients shall be tested using the same setup as in 4.5.4.4.5.1 (a), (b), and (c) except that the 0.04 volt output of the internal calibrator shall be connected to the vertical input and the height of the display shall be between 3.8 and 4.2 divisions.

(a) Set the variable frequency power supply to 115V, 48.5 Hz.
(b) Decrease the power frequency quickly to 47.0 Hz, hold for approximately 2 seconds, and return quickly to 48.5 Hz.
(c) The display shall return to normal if it momentarily went beyond specification.
(d) Repeat (b) four more times at 1-minute intervals.
(e) Repeat (a) through (d) with a starting frequency of 58.2 Hz and a transient frequency of 56.4 Hz.
(f) Repeat (a) through (d) with a starting frequency of 61.8 Hz and a transient frequency of 63.6 Hz.
(g) Repeat (a) through (d) with a starting frequency of 432 Hz and a transient frequency of 444 Hz.

4.5.4.4.5.1.2 Voltage transients.- Voltage transients shall be tested as follows:

(a) Connect the power cable of the equipment to the variac. The internal calibrator shall still be connected to the vertical input, giving a vertical display of between 3.8 and 4.2 divisions.
(b) Set the output voltage of the variac to 126.5V, 60 Hz.
(c) Quickly increase the output voltage of the variac to 151.8V, hold for approximately 2 seconds, and quickly return to 126.5V.
(d) The display shall return to normal if it momentarily went beyond specifications.
(e) Repeat (c) four more times at 2-minute intervals.
(f) Set the output voltage of the variac to 103.5V.
(g) Quickly decrease the output voltage of the variac to 82.8V, hold for approximately 2 seconds, and quickly return to 103.5V.
(h) The display shall return to normal if it momentarily went beyond specifications.
(i) Repeat (c) four more times at 2-minute intervals.

4.5.4.4.5.2 Ground test.-

(a) Connect the power cord of the equipment to the secondary of the 1.1 isolation transformer. Connect the primary of the transformer to the power line.
(b) Determine that one side of the power line is grounded.
(c) Connect the 2-volt output of the audio generator to the vertical INPUT of the equipment.
(d) Set the generator frequency and the sweep time for 1 kHz and a display of several cycles of the generator output.
(e) Set the equipment vertical sensitivity at 0.5V/div.
(f) Ground one side of the isolation transformer secondary; remove the ground and ground the other side.
(g) Reverse the power connections to the primary of the transformer. Repeat (f).
(h) No change in the display shall be observable under any of the grounded or ungrounded conditions specified in (a) through (g).

4.5.4.4.5.3 Input power.- The input power shall be checked as follows:

(a) Connect the power cord of the oscilloscope to the receptacle on the wattmeter. Connect the power cable of the wattmeter to 115V, 60 Hz power source.
(b) Read the input power to equipment from the wattmeter.
(c) The input power shall not exceed 50 watts.

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4.5.4.4.6 Miscellaneous.- Unless otherwise specified herein, items such as writing rate, weight, and size shall be measured with the operating controls of the plug-in preamplifier set as specified in 4.5.3.2.2.

4.5.4.4.6.1 Writing rate.- The writing rate shall be measured as follows:

(a) Connect the output of the pulse generator to the vertical INPUT of the equipment using a BNC to BNC cable and a N to BNC adapter. Connect the output of the square wave generator to the sync input of the pulse generator using a GR to BNC cable. Set the frequency of the square wave generator to 1 kHz. Set the controls of the pulse generator for + external sync, pulse length to 4 μsec, the output polarity to +, and the attenuator to 40 dB.

(b) Set the equipment controls as follows:

Volts/div	0.2
Time/div	1 μsec
TRIGGER SLOPE	+INT
TRIGGER MODE	SWEEP

Adjust the amplitude control of the pulse generator to give 5 divisions of vertical deflection on the equipment display. Adjust the stability controls for a stable display.

(c) Reduce the frequency of the square wave generator to 10 Hz. The pulse waveform shall remain visible on the display. (It may be necessary to cup hands around the bezel in order to shield the display from ambient light.) Adjustment of the intensity control shall provide both greater than and less than optimum intensity.

4.5.4.4.6.2 Weight and size.- Weight of the instrument with all accessories shall not exceed 26 pounds.

4.5.4.4.6.2.1 The overall dimensions of the equipment, excluding cover, handle, feet and latches, shall not exceed the following limits: 6-27/32 inches high and 8-3/4 inches wide by 14 inches deep. The overall dimensions of the equipment including cover, handle, feet and latches shall not exceed 8-7/32 inches high by 9-3/4 inches wide by 20-1/2 inches deep.

4.5.5 Environment inspection.- Environmental tests shall be conducted on each lot, except that the inspector may, at his discretion, waive this requirement in the light of evidence that the contractor's production and quality control are such as to assure the manufacture of satisfactory equipment. These tests shall comprise such examinations and testing as will prove satisfactory performance during or after severe environmental conditions of immersion, shock, vibration, inclination, temperature humidity, and salt spray. They shall also demonstrate durability and reliability.

4.5.5.1 Sampling plan.- These tests may be carried out concurrently with product control inspection on different samples from the lot being tested. The sample size and lot acceptance level shall be based upon table I. The Government inspector may, at his discretion, further reduce sample sizes.

TABLE I. Sampling for environmental inspection.

Lot size	Sample size	Number of units nonconforming in any environmental test	
		Acceptance number	Rejection number
1 to 100	1	0	1
101 to 300	2	0	1
301 to 500	4	1	2
501 or more	8	2	3

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4.5.5.1.1 Rejection.- Each of the sample units selected in accordance with table I shall be subjected to the tests of 4.5.5. The results of each test shall be compared with the requirements of this specification. Failure to conform to this specification for any such test shall be counted as a defect, and the unit shall be rejected. If the number of rejections in any sample exceed the acceptance number for the sample, the lot represented by the sample shall be rejected.

4.5.5.1.2 Noncompliance.- If a sample unit fails environmental inspections, the supplier shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of products which were manufactured with the same conditions, materials, processes, and so forth. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the supplier cannot determine the cause of failure, the matter shall be referred to the contracting officer.

4.5.5.1.3 Reinspection of conforming environmental inspection sample units.- Unless otherwise specified, sample units which have been subjected to and passed environmental inspections may be accepted on contract, provided that they are resubjected to and pass product inspection after repair of all visible damage.

4.5.5.2 Test equipment.- Test equipment for environmental tests shall be as specified herein and in MIL-E-16400.

4.5.5.3 Specific environmental tests.- In addition to the tests described below, the inspector may require performance of any of the tests of 4.5.4 or 4.5.5 as part of the temperature, altitude, humidity, shock, vibration, and life tests described herein.

4.5.5.3.1 Satisfactory operation tests.- The check for satisfactory operation during the course of environmental tests shall consist of the repetition of the following accuracy tests. When the equipment is operated in a controlled environment chamber, input and power connections shall be made through sealed feedthrough connections in the chamber and the equipment shall be energized 5 minutes before performing the set of tests of 4.5.5.3.1 or 4.5.5.3.3.2.

4.5.5.3.1.1 Voltage calibrator tests.-

(a) Connect output of the voltmeter calibrator to the vertical input A-AC of the unit under test. Set the calibration voltage to 400 mV peak-to-peak, 400 Hz ac. Set the vertical VOLTS/DIV switch to 0.1 volt/div. Adjust front panel controls for a stable presentation of the waveform and an amplitude of 4 divisions.

(b) Disconnect the voltage calibrator and connect the calibrator output of the oscilloscope to the vertical input A-AC. Set calibrator switch to 0.4 volt.

(c) The vertical deflection shall be between 3.80 and 4.20 divisions over temperature and humidity ranges I and II and shall be between 3.60 and 4.40 divisions over temperature and humidity range III.

4.5.5.3.1.2 Vertical accuracy test.-

(a) Connect output of the voltage calibrator to the vertical input A-AC of the unit. Set the calibration voltage to 50 mV peak-to-peak, 400 Hz ac. Set vertical VOLT/DIV switch to 0.01 volt/div.

(b) The vertical deflection shall be between 4.75 and 5.25 divisions over temperature and humidity ranges I and II and shall be between 4.50 and 5.50 divisions over temperature and humidity range III.

(c) Repeat for each vertical input when a dual channel preamplifier is used.

4.5.5.3.1.3 Horizontal accuracy test.-

(a) Connect the voltage calibrator to the external horizontal input of the unit. Set the calibration voltage to 2.5V, 400 Hz ac. Set the HORIZ MODE switch to X1 position.

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(b) The horizontal deflection shall be between 4.75 and 5.25 divisions over temperature and humidity ranges I and II and shall be between 4.50 and 5.50 divisions over temperature and humidity range III.

4.5.5.3.1.4 Sweep linearity test.-

(a) Connect the marker output of the time mark generator to the vertical input A-AC of the unit under test. Select the 0.1 second marker of the time mark generator. Set the sweep time switch of the unit to 0.1 second/div. Adjust front panel controls for a stable presentation.
(b) Align the first marker with the first graticule marking at the left of the screen.
(c) The eleventh marker shall be between 9.7 and 10.3 divisions from the first over temperature and humidity range I; between 9.50 and 10.50 divisions from the first over temperature and humidity range II; and between 9.20 and 10.80 divisions from the first over temperature and humidity range III.
(d) Repeat this test, using 1 msec marker and sweep time switch set for 1 msec/div and using 1 μsec marker with sweep time switch set for 1 μsec/div.

4.5.5.3.1.5 Expanded sweep linearity test.-

(a) Select 1 msec marker from time mark generator, set the sweep time switch to 1 msec/div and align the first marker with the first graticule line on the left. Set horizontal mode switch of the unit under test to 5X. There shall be three markers visible on the screen. Align the first marker with the first vertical line on the left of the screen.
(b) The third marker shall be between 9.50 and 10.50 divisions from the first over temperature and humidity range I and shall be between 9.00 and 11.00 divisions from the first over temperature humidity ranges II and III.

4.5.5.3.1.6 Bandpass test.-

(a) Using test setup of 4.5.4.4.2.1.1, increase the frequency of the signal generator, keeping the output amplitude constant to 100 kHz, 500 kHz, and 5 MHz.
(b) The amplitude of the deflection shall not exceed 8 divisions at 100 kHz and 500 kHz, and shall be equal to or greater than 5.6 divisions at 5 MHz.

4.5.5.3.1.7 Control settings.- Unless otherwise specified herein, the operating controls of the instrument shall be set as follows during the environmental tests:

Power	ON
Horizontal mode	SWEEP
Time/div	0.2. MS/DIV
Variable time/div	CAL
Trigger level	O
Trigger slope	INT
Stability	PRESET
Horizontal position	Adjust to center display
Focus	Adjust for best display
Intensity	Adjust for normal brightness
Scale illumination	Adjust for best display
Astigmatism	Adjust for best display

When using the high-gain preamplifier, set the preamplifier controls as follows:

Input selector	A-AC
Volts/div	0.01 volt per division
Variable volts/div	CAL
Vertical position	Center trace

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4.5.5.3.2 Temperature and humidity test.-

4.5.5.3.2.1 Nonoperating tests.- The equipment shall be subjected to the following temperature cycle and tests:

(a) Set up the equipment for test in a temperature-controlled chamber or room. Reduce chamber temperature in steps of 10° C each 30 minutes until -62° C is reached. Maintain -62° C +3° C for at least 24 hours.
(b) Increase chamber temperature in steps of 10° C each 30 minutes until +75° C is reached. Maintain +75° C +3°C for at least 4 hours.
(c) Reduce chamber temperature in steps of 10° C each 30 minutes until 25° C is reached. Maintain 25° C + 3° C for at least 4 hours.
(d) Perform the tests of 4.5.5.3.1. The accuracies of temperature and humidity (T and H) range I shall apply.

4.5.5.3.2.2 Operating test.- The equipment shall be capable of normal operation without alignment or adjustment of other than the operating controls throughout the following temperature cycle.

4.5.5.3.2.2.1 Conditioning.- Set up the equipment in a temperature- and humidity-controlled chamber or room. Energize the equipment. Maintain a temperature of 45° C +5° C and a relative humidity (RH) of 25 +5 percent for 8 hours. Decrease the temperature to 25° C in 2 equal steps of 30 minutes each. De-energize the equipment.

4.5.5.3.2.2.2 Cycling tests.- During the following tests, the relative humidity need not be controlled at temperatures below +5° C.

(a) Maintain 25° C +2° C and 25 percent +5 percent RH for 4 hours.
(b) Perform the tests of 4.5.5.3.1. The accuracies of T and H range I shall apply.
(c) Decrease the temperature to 15° C. Maintain 15° C + 2° C and 25 percent +5 percent RH for 4 hours.
(d) Perform the tests of 4.5.5.3.1. The accuracies of T and H range I shall apply.
(e) Decrease the temperature to 0° C in two approximately equal steps of 20 minutes each. Maintain 0° C +2° C and not more than 25 percent RH for 4 hours.
(f) Perform the tests of 4.5.5.3.1. The accuracies of T and H range II shall apply.
(g) Decrease the temperature to -28° C in three approximately equal steps of 30 minutes each. Maintain -28° C +2° C and not more than 25 percent RH for 4 hours.
(h) Perform the tests of 4.5.5.3.1. The accuracies of T and H range III shall apply.
(i) Increase the temperature to +35° C in steps of approximately 10° C each 30 minutes. Maintain 35° C +2° C and 25 percent +5 percent RH for 4 hours.
(j) Perform the tests of 4.5.5.3.1. The accuracies of T and H range I shall apply.
(k) Increase the temperature to +50° C in two equal steps of 20 minutes each. Maintain 50° C +2° C and 25 percent +5 percent RH for 4 hours.
(l) Perform the tests of 4.5.5.3.1. The accuracies of T and H range II shall apply.
(m) Increase the temperature to 65° C in two equal steps of 30 minutes each. Maintain 65° C +2° C and 25 percent +5 percent RH for 4 hours.
(n) Perform the tests of 4.5.5.3.1. The accuracies of T and H range III shall apply.
(o) Decrease the temperature to 25° C in steps of approximately 10° C each 30 minutes.
(p) Repeat (a) through (o) for relative humidities of 50 percent, 75 percent, 90 percent, and 95 percent. The accuracies shall be as required therein except as hereinafter specified:

RH	Temperature	Accuracy of T and H range
90 percent	0° C to +50° C	II
any	-28° C to 0° C	III
95 percent	any	III

(q) Repeat 4.5.5.3.2.2.1 and 4.5.5.3.2.2.2 (a) and (b).

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4.5.5.3.2.3 Examination.- The equipment shall be examined in detail to detect evidence of physical degradation, such as corrosion of metal parts, distortion of plastic parts, and insufficient lubrication of moving parts. When it is necessary to replace parts to obtain satisfactory performance of the equipment, the failed part or parts shall be analyzed to determine the cause of unsatisfactory operation. The results of the analysis shall be reported with the results of measurements of the equipment operating parameters. The unsatisfactory parts or material shall be replaced by adequate substitutes.

4.5.5.3.3 Vibration test.-

(a) Using suitable canvas straps, mount the equipment (installed in its combination case but with the cover separately mounted) to the vibration table with the major horizontal axis of the equipment horizontal and parallel to the direction of vibration.
(b) Energize the equipment. After a 30-minute warm-up period, perform the tests of 4.5.5.3.1. During vibration the equipment shall be measuring the satisfactory operation check of 4.5.5.3.3.2.
(c) Determine the presence of resonances in the equipment under test by vibrating at frequencies from 5 Hz to 33 Hz at a table vibratory amplitude (for example, one-half the total excursion) of 0.010 ±0.002 inch. The change in frequency shall be made in discrete steps of 1 Hz and maintained for about 15 seconds. Record frequencies at which resonances occur.
(d) Repeat (c) with 5-minute steps.
(e) Perform the tests of 4.5.5.3.1.
(f) Vibrate the equipment for a total period of at least 2 hours at the resonant frequencies chosen by the test engineer. If no resonance was observed, this test shall be performed at 33 Hz. The amplitudes of vibration shall be in accordance with table III.

TABLE III - Amplitudes of vibration.

Frequency range Hz	Table amplitude plus or minus inch
5 to 15	0.030 ±0.006
16 to 25	.020 ±.004
26 to 33	.010 ±.002

(g) Perform the tests of 4.5.5.3.1.
(h) Repeat (a) through (f) with the equipment mounted with its major horizontal axis parallel to the table and perpendicular to the direction of vibration.
(i) Repeat (a) through (f) with the equipment mounted with its major horizontal axis perpendicular to the table.
(j) Acceptability will be contingent upon the ability of the equipment to withstand tests specified above and the ability to perform its principal functions during and after vibration tests. Minor damage or distortion will be permitted during test providing such damage or distortion does not in any way impair the ability of the equipment to perform its principal functions. Non-repetitive failures of such parts as vacuum tubes, condensers, and wiring, which can be easily replaced or repaired, are generally considered minor failures. If minor failures occur, the test shall be stopped, the repair made, and the entire test in that plane repeated. If major failures occur, the test shall be discontinued and, after repair, the entire test shall be repeated, unless otherwise directed by the command or agency concerned. The manufacturer may, at his option, substitute an entirely new equipment for retest. If this option is taken, it shall be noted in the test report furnished in accordance with 4.5.5.3.3.1.

4.5.5.3.3.1 Test report.- The test report (see 6.3) to be furnished the command or agency concerned by the testing laboratory shall include detailed descriptions of any damage or malfunctioning incurred and at what stage in the tests it occurred. Photographs of physical damage shall be included. Recommendations are desired as to what corrective measure, if any, should be taken. At the discretion of the test engineer, it shall also

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include other pertinent information, such as the overall dimensions of the equipment, its weight, approximate location of the center of gravity, and a sketch or photograph of the methods in mounting it on the test machines.

4.5.5.3.3.2 Satisfactory operation check.- Unless otherwise specified herein, the following tests shall be performed during the environmental inspection:

(a) Set calibrator switch of the unit under test to 0.04 volt position. Connect the calibrator output to vertical A-AC input: set V/div switch to 0.001 V/div. There should be 4 complete square waves with 4 division amplitude. .
(b) The unit shall be considered to be operating satisfactorily if operation of the front panel control permits an acceptable presentation of the display.

4.5.5.3.4 Shock test.- Unless otherwise specified herein, each equipment shall be subject to the shock test, grade A, class I, type A for lightweight equipment in accordance with MIL-S-901. The following conditions shall apply:

(a) The equipment shall be mounted with canvas straps satisfactory to the command or agency concerned. The cover shall be removed from the case and separately mounted to the shock table.
(b) The equipment shall be energized for one-half hour. The tests of 4.5.5.3.1 shall be performed before any blows and after each blow.
(c) During the blows, the equipment shall be measuring the satisfactory operation test of 4.5.5.3.3.2.
(d) Three blows shall be applied parallel to each of three principal axes of the apparatus being tested. The three blows in each horizontal direction shall be with heights of hammer drop of 1 foot, 2 feet, and 3 feet, respectively, and the three blows in the vertical direction shall be with heights of 2 feet, 3 feet, and 4 feet, respectively.

4.5.5.3.5 Accelerated life tests.- The equipment shall be subjected to the following conditioning and tests.

4.5.5.3.5.1 Conditioning.-

4.5.5.3.5.1.1 Normal operation.- The equipment shall first be set up in a chamber in an ambient temperature of 25° ±5°C with a relative humidity of 50 ±3 percent, and operated under normal conditions of line voltage and frequency in accordance with 3.5.1. Following warm-up, the tests of 4.5.5.3.1 shall be performed. The results shall be recorded as reference data.

4.5.5.3.5.1.2 High-temperature operation.- Chamber temperature shall be increased to 65° ±5° C and not less than 90 percent relative humidity, and the equipment shall be operated under nominal supply voltage and frequency for a period of 2 hours. During the last 30 minutes the tests of 4.5.5.3.1 shall be performed.

4.5.5.3.5.2 Test cycling.- Immediately following the high-temperature conditioning, the equipment shall be subjected to the following series of cycles while operating in an ambient temperature of 65° ±5° C and a relative humidity of not less than 90 percent.

(a) Increase input voltage to 126.5 volts and operate in this condition for 1 hour and 15 minutes.
(b) Decrease input voltage to 103.5 volts and operate in this condition for 1 hour.
(c) Increase input voltage to 115 volts, operate the equipment for 30 minutes in this condition, and then remove power for 15 minutes. During the last 30 minutes of operation repeat 4.5.5.3.1.
(d) At the end of the 15-minute shut-down period, energize the equipment with 103.5 volts and operate in this condition for 1 hour.
(e) Increase input voltages to 126.5 volts and operate in this condition for 1 hour and 15 minutes.
(f) Decrease input voltage to nominal and operate for 30 minutes repeating 4.5.5.3.1. At the end of the 30-minute period, shut down the equipment for 15 minutes.
(g) Repeat (a) through (f) without interruption for a period of 200 hours, performing the tests of 4.5.5.3.1 on every fourth cycle thereafter.

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4.5.5.3.5.3 Procedure in case of shutdown.- Should a single shutdown for more than 1 hour or a total down time of 5 hours during the 200 hours of continuous operation be necessary, the continuation of the test shall be for 200 hours from the time of resumption of the run.

4.5.5.3.5.4 Temperature measurement.- Temperature measuring instruments shall be placed at critical points throughout the equipment, covering suspected hot spots. Data shall be recorded at least once during each cycle, at the end of the period of high voltage operation.

4.5.5.3.6 Dripproof (enclosure) test for combination case.-

(a) Repeat 4.5.5.3.1.
(b) Mount the equipment de-energized in its combination case with cover installed in such fashion as to permit tilting of its vertical axis in any direction to the extent of 75 degrees to the horizontal. The equipment so mounted shall be placed on the tilt table directly below a hose with a 1-inch uniform diameter nozzle installed not less than 3 feet above the top of the equipment. The nozzle shall be fitted with a domestic type sprinkling faucet.
(c) The enclosure of the equipment shall be tested by subjecting it to a stream of water under a velocity head at the nozzle of 1 foot. The nozzle shall be directed downward to the test position.
(d) During the test the unit shall be tilted in all directions and subjected to the test in accordance with the following schedule:
    (1) Stream directed on top (with table horizontal) for 15 minutes.
    (2) Stream directed for 5 minutes onto each completely closed side and each side containing controls, panels, cable entrances, louvers, and so forth (tilt the table so that the side being sprayed is 75 degrees from the horizontal).
(e) Return the equipment to the horizontal (tilt table horizontal) and repeat 4.5.5.3.1. Remove the instrument from its combination case. Any leakage accumulated in pockets in the bottom or in any other part of the enclosure will constitute failure of this test.

4.5.5.3.7 Altitude test.-

(a) Place the equipment in a suitable altitude chamber at 25° ±2° C.
(b) Perform the tests of 4.5.5.3.1. The accuracies of T and H range I shall apply.
(c) During the altitude test, the equipment shall be measuring the satisfactory operation check of 4.5.5.3.3.2.
(d) Reduce the pressure to 9 inches of mercury. Maintain for 15 minutes.
(e) Observe the reading on the equipment under test. There should be no change except for allowable drift.
(f) Turn equipment off. Reduce pressure to 3.4 inches of mercury. Maintain for 15 minutes.
(g) Increase the pressure to normal ambient atmosphere, turn equipment on, and perform the tests of 4.5.5.3.1. The accuracies of T and H range I shall apply.

4.5.5.3.8 Post environmental tests.- The specific operating tests of 4.5.5.3.1 shall be repeated after all of the environmental tests of 4.5.5.3.2 through 4.5.5.3.7 being performed on a given unit have been performed. Any unit failing to conform to the requirements of 4.5.5.3.1 shall be counted as failing to pass the environmental requirements and shall be rejected.

4.5.5.3.9 Salt spray test.- Parts of the equipment not in accordance with MIL-E-16400 shall be subjected to a 100-hour salt spray test in accordance with FED. TEST METHOD STD. No. 151 as specified in MIL-E-16400.

4.5.5.3.10 Electromagnetic interference and undesired radiation tests.- The following electromagnetic interference and undesired radiation tests shall be conducted in accordance with MIL-STD-462:

CE01 RE01
CE03 RE02

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CS01	RS01
CS02	RS03
CS06	

4.5.5.3.11 Weld test.- When required by the inspector, sample pieces representative of the product, and welded on the machines used therefore, shall be tested to destruction to determine conformance with the welding requirements of MIL-E-16400.

4.6 Materials inspection.- Materials inspection shall be as specified in MIL-E-16400.

4.7 Inspection of preparation for delivery.- Inspection of preparation for delivery shall be conducted to insure conformance with the requirements of Section 5.

4.8 Retention of qualification.- To retain qualification, the supplier shall forward a report at 24 month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of:

(a) A summary of the results of the tests performed for inspection of product for delivery, Models A, B, C, and D, indicating as a minimum the number of lots that have passed and the number that have failed. The results of tests of all reworked lots shall be identified and accounted for.

(b) A summary of the results of tests performed for qualification verification inspection, Models A, B, C, and D, including the number and mode of any subgroup failures. The summary shall include results of all qualification verification inspection tests performed on completed lots during the 24 month period. If the summary of the test results indicates nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

Failure to submit the report within 30 days after the end of each 24 month period may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the supplier shall immediately notify the qualifying activity at any time that the inspection data indicates noncompliance of the product to meet the requirements of this specification.

In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during 2 consecutive reporting periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit the oscilloscopes to testing in accordance with the qualification inspection requirements to the extent required by the qualifying activity.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking.- The equipment accessories, technical publications, and repair parts shall be preserved and packaged by level A or C; packed by level A, B, or C, and marked, as specified in the contract or order (see 6.2), in accordance with the requirements of MIL-P-17555. Method II shall apply for level A packaging of equipment, except that Method 1C-5 shall apply where the storage time before use will not exceed 6 months.

6. NOTES

6.1 Intended use.- The equipment described herein is intended for Military service aboard ships, planes, and in shore facilities.

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6.2 Ordering data. -Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Cathode ray tube phosphor, if other than P31 phosphor (3.4.2).
(c) Dual-channel preamplifier, if required (3.4.3.3).
(d) Dust cover, when required (3.6.1.4.1).
(e) Marking, levels of preservation and packaging and packing required (5.1).

6.3 Contract data requirements. -Data generated by this document are not deliverable unless specified on the Contract Data Requirements List (DD 1423). The data required by this specification include, but are not restricted to, the following:

(a) Request for approval of parts employed for reduction of electromagnetic interference (3.5.3).
(b) Request for approval of parts and materials containing mercury (3.8).
(c) A written description of contractor's quality assurance plan, including flow charts (4.1.1).
(d) Three (3) sets of information on standard and non-standard parts (4.3.1).
(e) Test reports, including detailed description of any damage or malfunction and photographs of physical damage and method of mounting on test machine (4.5.5.3.3.1).
(f) Retention of qualification reports (4.8).
(g) Technical manuals, including operating instructions and transformer diagram, as required in the contract or order.

6.4 With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List (QPL 23124) whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Electronic Systems Command, Department of the Navy, Washington, D.C. 20360, and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.5).

6.5 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.6 CHANGES FROM PREVIOUS ISSUE.- THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, AND SO FORTH) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Custodians:
Army - EL
Navy - EC
Air Force -

Review activities:
Army - MI
Navy - OS, SH, AS
Air Force -

User activities:
Army - WC, AV
Navy - MC, YD, CG
Air Force -

Preparing activity:
Navy - EC

(Project 6625-0332)

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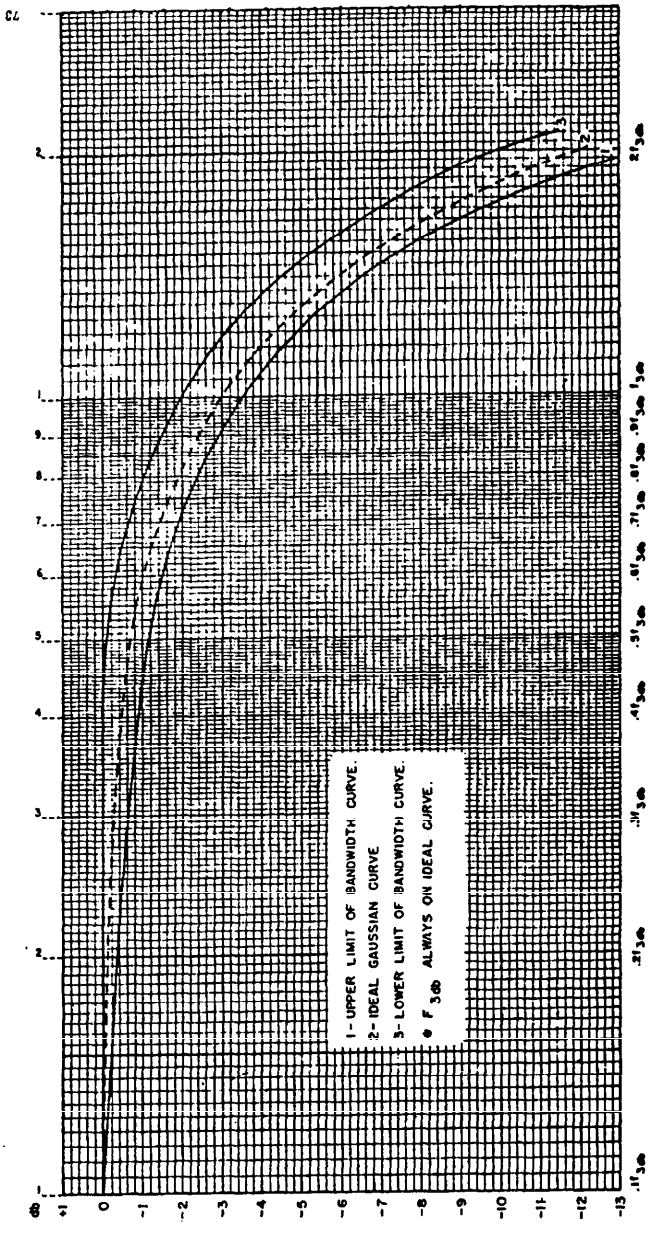


FIGURE 1. Bandpass curve.

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CIRCUIT SYMBOL DESIGNATION	DESCRIPTION	PART NUMBER	VALUE	CRITICAL TOLERANCE

FIGURE 2. Format for information on standard and nonstandard parts.

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MAXIMUM USABLE SIZE WITHOUT REDESIGN					CHECK STANDARD PART PER MIL-STD-242, 200 or 701	CHECK IF NON-STANDARD PART	QUALITY ASSURANCE		
DIAMETER	LENGTH	HEIGHT	WIDTH OR THICKNESS	WEIGHT			PER SPEC. NO.	EXCEPT FOR	OTHER THAN MIL SPEC.

FIGURE 2. Format for information on standard and nonstandard parts (cont'd.)

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CHECK IF NON-STANDARD PART PER MIL-STD-242, 200, or 701	PER MIL SPEC CITED IN MIL-STD-242, 200, or 701 (State which Spec applies)	PER OTHER MIL SPEC (State which Spec applies)	STANDARD PART EXCEPT FOR (List exceptions: size, value, tolerance, etc.)	PER MIL SPEC CITED IN MIL-STD-242, except for (List exceptions)	PER OTHER MIL SPEC EXCEPT FOR (List exception)	CHECK IF NOT PER A MIL-STD OR SPEC WITH OR WITHOUT EXCEPTION

FIGURE 2. Format for information on standard and nonstandard parts (cont'd.)

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PREVIOUSLY APPROVED FOR		
EQUIPMENT	NObsr	MANUFACTURER OR QPL AND PART NUMBER

FIGURE 2. Format for information on standard and nonstandard parts (cont'd.)

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LIST ANY STANDARD PART WHICH CAN BE USED WITHOUT AFFECTING CIRCUIT PERFORMANCE	CHECK IF STANDARD PART REQUIRES MECHANICAL REDESIGN	CHECK IF STANDARD PART REQUIRES ELECTRICAL REDESIGN	CHECK IF STANDARD PART IS INTERCHANGEABLE WITH PROPOSED PART	CHECK IF STANDARD PART IS CONTRACTUALLY REQUIRED	EXPLANATORY NOTES

FIGURE 2. Format for information on standard and nonstandard parts (cont'd.)

Source: https://assist.dla.mil -- Downloaded: 20261O-19T15:36Z
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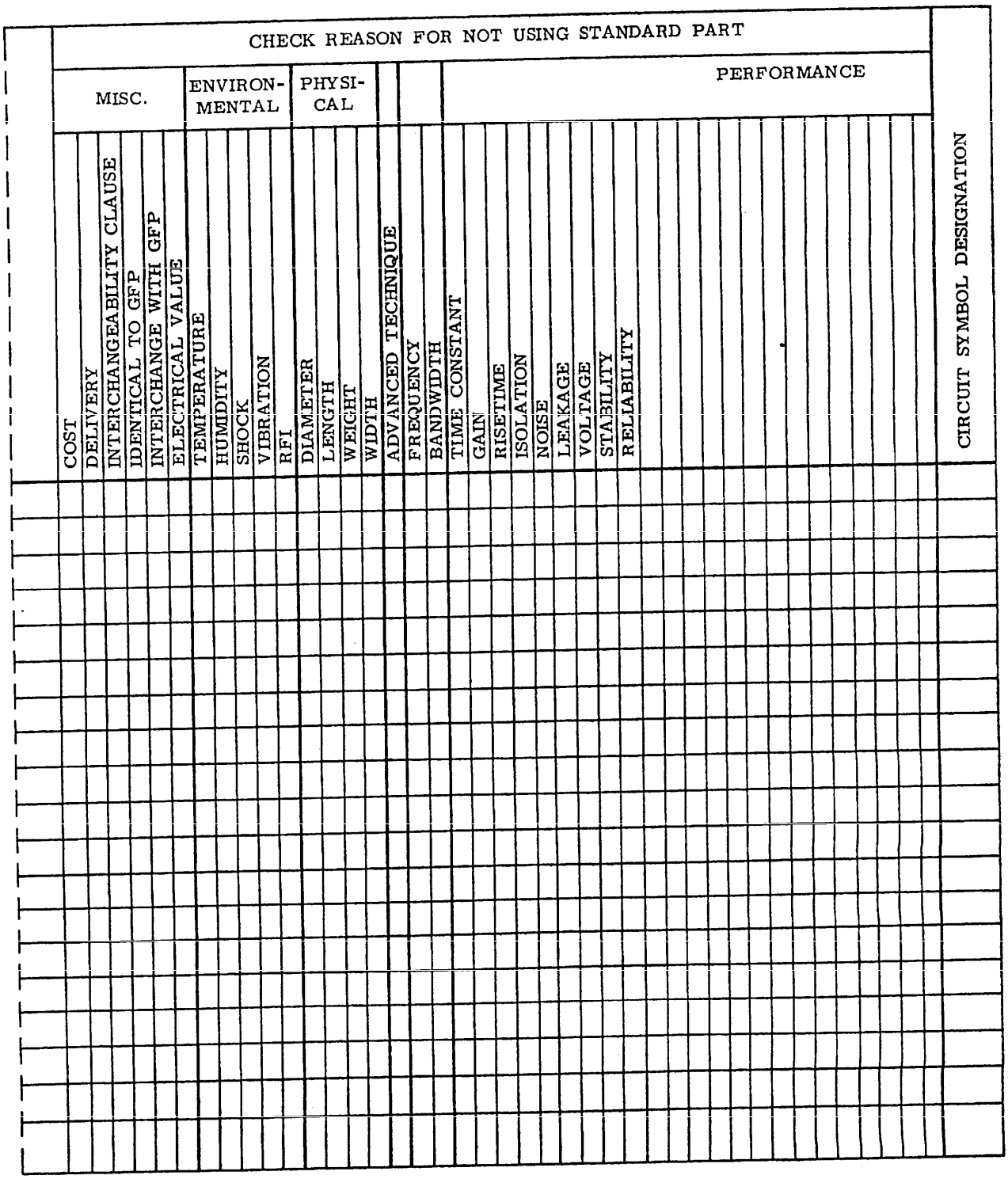


FIGURE 2. Format for information on standard and nonstandard parts (cont'd.)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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NAVY DEPARTMENT

OFFICIAL BUSINESS

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

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